University of the Philippines - Baguio

Archie Junio's Reference Notebook

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1. Data Structures

1.1. Segment Tree.

```
class SegmentTree: ------------------------------------------------------
--- def __init__(self, data): ------------------------------------------
------- self.n = len(data) -------------------------------------------
------- self.tree = [0] * (2 * self.n) -------------------------------
------- for i in range(self.n): --------------------------------------
----------- self.tree[self.n + i] = data[i] --------------------------
------- for i in range(self.n - 1, 0, -1): ---------------------------
----------- self.tree[i] = self.tree[i << 1] + self.tree[i << 1 | 1] -
 ------------------------------------------------------------------------
--- def update(self, p, value): ----------------------------------------
------- p += self.n --------------------------------------------------
------- self.tree[p] = value -----------------------------------------
------- while p > 1: -------------------------------------------------
----------- self.tree[p >> 1] = self.tree[p] + self.tree[p ^ 1] ------
----------- p >>= 1 ------------------------------------------------
 ------------------------------------------------------------------------
--- def query(self, l, r): ---------------------------------------------
------- res = 0 ------------------------------------------------------
------- l += self.n --------------------------------------------------
------- r += self.n --------------------------------------------------
------- while l < r: -------------------------------------------------
----------- if l & 1: ----------------------------------------------
--------------- res += self.tree[l] ------------------------------
--------------- l += 1 -------------------------------------------
----------- if r & 1: ----------------------------------------------
--------------- r -= 1 -------------------------------------------
--------------- res += self.tree[r] ------------------------------
----------- l >>= 1 ------------------------------------------------
----------- r >>= 1 ------------------------------------------------
------- return res ---------------------------------------------------
```

1.2. Union Find.

```
class DSU: --------------------------------------------------------------
--- def __init__(self, n): ---------------------------------------------
```

```
        self.parent = list(range(n))
        self.size = [1] * n

    def find(self, i):
        if self.parent[i] == i:
            return i
        self.parent[i] = self.find(self.parent[i])
        return self.parent[i]

    def union(self, i, j):
        root_i = self.find(i)
        root_j = self.find(j)

        if root_i != root_j:
            if self.size[root_i] < self.size[root_j]:
                root_i, root_j = root_j, root_i
            self.parent[root_j] = root_i
            self.size[root_i] += self.size[root_j]
            return True
        return False

    def get_size(self, i):
        return self.size[self.find(i)]
```

1.3. 2D Sparse Table.

```
import math

class SparseTable2D:
    def __init__(self, A):
        self.A = A
        self.n = len(A)
        self.m = len(A[0])

        self.LOGN = int(math.log2(self.n)) + 1
        self.LOGM = int(math.log2(self.m)) + 1

        self.lg = [0] * (max(self.n, self.m) + 1)
        for i in range(2, len(self.lg)):
            self.lg[i] = self.lg[i // 2] + 1

        self.st = [[[[0] * self.m for _ in range(self.n)]
                    for _ in range(self.LOGM)]
                   for _ in range(self.LOGN)]

        self.build()

    def build(self):
        for i in range(self.n):
            for j in range(self.m):
                self.st[0][0][i][j] = self.A[i][j]

        for bj in range(1, self.LOGM):
            for i in range(self.n):
                for j in range(self.m - (1 << bj) + 1):
                    self.st[0][bj][i][j] = max(
                        self.st[0][bj - 1][i][j],
                        self.st[0][bj - 1][i][j + (1 << (bj - 1))]
                    )

        for bi in range(1, self.LOGN):
            for i in range(self.n - (1 << bi) + 1):
                for j in range(self.m):
                    self.st[bi][0][i][j] = max(
                        self.st[bi - 1][0][i][j],
                        self.st[bi - 1][0][i + (1 << (bi - 1))][j]
                    )

        for bi in range(1, self.LOGN):
            for bj in range(1, self.LOGM):
                for i in range(self.n - (1 << bi) + 1):
                    for j in range(self.m - (1 << bj) + 1):
                        self.st[bi][bj][i][j] = max(
                            self.st[bi - 1][bj - 1][i][j],
                            self.st[bi - 1][bj - 1][i + (1 << (bi - 1))][j],
                            self.st[bi - 1][bj - 1][i][j + (1 << (bj - 1))],
                            self.st[bi - 1][bj - 1][i + (1 << (bi - 1))][j + (1 << (bj - 1))]
                        )

    def query(self, x1, y1, x2, y2):
        kx = self.lg[x2 - x1 + 1]
        ky = self.lg[y2 - y1 + 1]

        return max(
            self.st[kx][ky][x1][y1],
            self.st[kx][ky][x2 - (1 << kx) + 1][y1],
            self.st[kx][ky][x1][y2 - (1 << ky) + 1],
            self.st[kx][ky][x2 - (1 << kx) + 1][y2 - (1 << ky) + 1]
        )
# Input Format: st = SparseTable2D(grid)
#               print(st.query(1, 1, 2, 2))
```

2. Dynamic Programming

2.1. Lichao Tree.

```
class Line:
    __slots__ = ['m', 'c']
    def __init__(self, m=0, c=float('-inf')):
        self.m = m
        self.c = c

    def eval(self, x):
        return self.m * x + self.c

class Node:
    __slots__ = ['line', 'left', 'right']
    def __init__(self):
        self.line = Line()
        self.left = None
        self.right = None

class LiChaoTree:
    def __init__(self, min_x=-10**12, max_x=10**12):
        self.root = Node()
        self.min_x = min_x
        self.max_x = max_x

    def add_line(self, m, c):
        self._insert(self.root, self.min_x, self.max_x, Line(m, c))

    def _insert(self, node, l, r, new_line):
        mid = (l + r) // 2

        left_better = new_line.eval(l) > node.line.eval(l)
        mid_better = new_line.eval(mid) > node.line.eval(mid)

        if mid_better:
            node.line, new_line = new_line, node.line
        if r - l <= 1:
            return

        if left_better != mid_better:
            if not node.left:
                node.left = Node()
            self._insert(node.left, l, mid, new_line)
        else:
            if not node.right:
                node.right = Node()
            self._insert(node.right, mid, r, new_line)

    def query(self, x):
        return self._query(self.root, self.min_x, self.max_x, x)

    def _query(self, node, l, r, x):
        if not node:
            return float('-inf')
        if r - l <= 1:
            return node.line.eval(x)

        mid = (l + r) // 2
        res = node.line.eval(x)

        if x < mid:
            res = max(res, self._query(node.left, l, mid, x))
        else:
            res = max(res, self._query(node.right, mid, r, x))

        return res
```

2.2. Divide And Conquer.

```
INF = float('inf')

def solve_dp(g, k_L, k_R, n_L, n_R, dp, cost_func):
    if n_L > n_R:
        return

    n_M = (n_L + n_R) // 2
    best_k = -1
    dp[g][n_M] = INF

    for k in range(k_L, min(n_M, k_R) + 1):
        current_val = dp[g - 1][k] + cost_func(k + 1, n_M)
        if current_val < dp[g][n_M]:
            dp[g][n_M] = current_val
            best_k = k

    solve_dp(g, k_L, best_k, n_L, n_M - 1, dp, cost_func)
    solve_dp(g, best_k, k_R, n_M + 1, n_R, dp, cost_func)
```

3. Geometry

3.1. Angles And Rotations.

```
def angle(a, b, c): # angle formed by abc in radians: PI < x <= PI
    return abs((arg(a-b) - arg(c-b)) % (2 * math.pi))
def rotate(p, a, d): # rotate point a about pivot p CCW at d radians
    return p + (a - p) * complex(math.cos(d), math.sin(d))
```

3.2. Distances.

```
# Distance: Point to 2D Line (ax + by + c = 0)
def dist_pt_line_eq(p, a, b, c):
    return abs(a*p.real + b*p.imag + c) / math.hypot(a, b)

# Distance: Point to 2D Line through points a and b
def dist_pt_line(p, a, b):
    cp = (p - a).conjugate() * (b - a)
    return abs(cp.imag) / abs(b - a)

# Distance: Point to 3D Plane (ax + by + cz + d = 0)
def dist_pt_plane(p, a, b, c, d):
    # math.hypot supports 3+ arguments in Python 3.8+
    return abs(a*p.x + b*p.y + c*p.z + d) / math.hypot(a,b,c)

# Distance: Shortest distance between 3D lines AB and CD
def dist_line_3d(A, B, C, D):
    u, v, w = B - A, D - C, A - C
    a, b = u.dot(u), u.dot(v)
    c, d, e = v.dot(v), u.dot(w), v.dot(w)
    det = a * c - b * b
```

```
--- if det < 1e-9: # Parallel lines ----------------------------------
------- s, t = 0.0, d / b if b > c else e / c ----------------------
--- else: # Skew lines -----------------------------------------------
------- s = (b * e - c * d) / det ----------------------------------
------- t = (a * e - b * d) / det ----------------------------------
--- p1 = P3(A.x + u.x*s, A.y + u.y*s, A.z + u.z*s) -------------------
--- p2 = P3(C.x + v.x*t, C.y + v.y*t, C.z + v.z*t) -------------------
--- vec = p1 - p2 ----------------------------------------------------
--- return math.sqrt(vec.dot(vec)) -----------------------------------
```

3.3. Intersections.

3.3.1. *Circle Line Intersection.*

```
EPS = 1e-9 -------------------------------------------------------------
def projLine(p, a, b): # Project point p onto line segment ab --------
--- v = b - a --------------------------------------------------------
--- scalar = ((p - a) * v.conjugate()).real / abs(v)**2 -------------
--- return a + v * scalar --------------------------------------------
 ----------------------------------------------------------------------
def rotate(c, p, t): # Rotate point p around center c by t radians ---
--- return c + (p - c) * cmath.rect(1, t) ----------------------------
 ----------------------------------------------------------------------
def CL_inter(c, r, a, b): ----------------------------------------------
--- p = projLine(c, a, b) --------------------------------------------
--- d = abs(c - p) ---------------------------------------------------
--- ans = [] ---------------------------------------------------------
---  ----------------------------------------------------------------
--- if d > r + EPS: pass # None --------------------------------------
--- elif d > r - EPS: ans.append(p) # Tangent ------------------------
--- elif d < EPS: # Diameter -----------------------------------------
------- v = r * (b - a) / abs(b - a) -------------------------------
------- ans.append(c + v) ------------------------------------------
------- ans.append(c - v) ------------------------------------------
--- else: # Secant ---------------------------------------------------
------- t = math.acos(d / r) ---------------------------------------
------- p = c + (p - c) * r / d ------------------------------------
------- ans.append(rotate(c, p, t)) --------------------------------
------- ans.append(rotate(c, p, -t)) -------------------------------
--- return ans -------------------------------------------------------
```

3.3.2. *Circle Circle Intersection.*

```
EPS = 1e-9 -------------------------------------------------------------
def rotate(c, p, t): ---------------------------------------------------
--- # Rotate point p around center c by t radians --------------------
--- return c + (p - c) * cmath.rect(1, t) ----------------------------
 ----------------------------------------------------------------------
def CC_inter(c1, r1, c2, r2): ------------------------------------------
--- d = abs(c1 - c2) -------------------------------------------------
--- ans = [] ---------------------------------------------------------
--- if d < EPS: # Concentric circles ---------------------------------
------- if abs(r1 - r2) < EPS: -------------------------------------
----------- pass # Infinite intersections (same circle) --------------
--- elif r1 < EPS: # Circle 1 is a point -----------------------------
------- if abs(d - r2) < EPS: --------------------------------------
----------- ans.append(c1) -------------------------------------------
--- else: # General case using Law of Cosines -------------------------
------- s = (r1*r1 + d*d - r2*r2) / (2*r1*d) -----------------------
------- s = max(-1.0, min(1.0, s)) # Prevent float domain errors -----
------- t = math.acos(s) -------------------------------------------
------- # Point on C1's radius pointing towards C2 -------------------
------- mid = c1 + (c2 - c1) * r1 / d ------------------------------
------- ans.append(rotate(c1, mid, t)) -----------------------------
------- if abs(math.sin(t)) >= EPS: --------------------------------
----------- # FIXED: Rotate around c1, not c2! ----------------------
----------- ans.append(rotate(c1, mid, -t)) --------------------------
--- return ans -------------------------------------------------------
```

3.3.3. *Line Intersect.*

```
def line_inter_fast(a, b, c, d): ---------------------------------------
--- A = cross(d - a, b - a) ------------------------------------------
--- B = cross(c - a, b - a) ------------------------------------------
--- return (B * d - A * c) / (B - A) if B - A else None --------------
```

3.4. Closest Pair.

```
def closest_pair(pts): -------------------------------------------------
--- if len(pts) <= 1: return float('inf') ----------------------------
--- px = sorted(pts) # Automatically sorts by x, then y --------------
--- def solve(l, r): # Base case: Brute force for small partitions ---
------- if r - l <= 3: ---------------------------------------------
----------- d = float('inf') -----------------------------------------
----------- for i in range(l, r): ------------------------------------
--------------- for j in range(i + 1, r): ------------------------------
------------------ d = min(d, math.dist(px[i], px[j])) ---------------
----------- return d -------------------------------------------------
------- mid = (l + r) // 2 -----------------------------------------
------- mid_x = px[mid][0] -----------------------------------------
------- # Recursively find the minimum in both halves ----------------
------- d = min(solve(l, mid), solve(mid, r)) ----------------------
------- # Gather points in the middle strip ---------------------------
------- strip = [p for p in px[l:r] if abs(p[0] - mid_x) < d] -------
------- strip.sort(key=lambda p: p[1]) # Sort strip by y -------------
------- # Check at most 7 geometric neighbors in the strip -----------
------- for i, p1 in enumerate(strip): -----------------------------
----------- for p2 in strip[i+1 : i+8]: ------------------------------
--------------- d = min(d, math.dist(p1, p2)) ---------------------------
------- return d ---------------------------------------------------
--- return solve(0, len(px)) -----------------------------------------
```

3.5. Polygon Centroid.

```
def cross(a, b): -------------------------------------------------------
--- return a[0] * b[1] - a[1] * b[0] ---------------------------------
 ----------------------------------------------------------------------
def centroid(points): --------------------------------------------------
--- n = len(points) --------------------------------------------------
--- ans_x, ans_y = 0.0, 0.0 ------------------------------------------
--- z = 0.0 ----------------------------------------------------------
 ----------------------------------------------------------------------
--- for i in range(n): -----------------------------------------------
------- j = (i - 1) % n --------------------------------------------
------- cp = cross(points[j], points[i]) ---------------------------
 ----------------------------------------------------------------------
------- ans_x += (points[j][0] + points[i][0]) * cp ----------------
------- ans_y += (points[j][1] + points[i][1]) * cp ----------------
------- z += cp ----------------------------------------------------
 ----------------------------------------------------------------------
--- return (ans_x / (3 * z), ans_y / (3 * z)) ------------------------
```

3.6. Convex Hull.

3.6.1. *Convex Hull.*

```
def cross(o, a, b): # 2D cross product of OA and OB vectors ----------
--- return (a[0]-o[0])*(b[1]-o[1]) - (a[1]-o[1])*(b[0]-o[0]) ---------
 ----------------------------------------------------------------------
def convex_hull(pts): # Sorts by x, then by y automatically ----------
--- pts = sorted(set(pts))  ------------------------------------------
--- if len(pts) <= 1: ------------------------------------------------
------- return pts -------------------------------------------------
--- def build(iterable): ---------------------------------------------
------- h = [] -----------------------------------------------------
------- for p in iterable: -----------------------------------------
----------- while len(h) >= 2 and cross(h[-2], h[-1], p) <= 0: -------
--------------- h.pop() --------------------------------------------
----------- h.append(p) ----------------------------------------------
------- return h ---------------------------------------------------
--- lower = build(pts) -----------------------------------------------
--- upper = build(reversed(pts)) -------------------------------------
--- return lower[:-1] + upper[:-1] -----------------------------------
```

3.6.2. *Convex Hull 3D.*

```
class P3: --------------------------------------------------------------
--- __slots__ = ['x', 'y', 'z'] --------------------------------------
--- def __init__(self, x=0, y=0, z=0): -------------------------------
------- self.x, self.y, self.z = x, y, z ---------------------------
--- def __sub__(self, o): --------------------------------------------
------- return P3(self.x-o.x, self.y-o.y, self.z-o.z) --------------
--- def cross(self, o): ----------------------------------------------
------- return P3(self.y*o.z - self.z*o.y, -------------------------
----------------- self.z*o.x - self.x*o.z, -------------------------
----------------- self.x*o.y - self.y*o.x) -------------------------
--- def dot(self, o): ------------------------------------------------
------- return self.x*o.x + self.y*o.y + self.z*o.z ----------------
--- def __eq__(self, o): ---------------------------------------------
------- return (self.x, self.y, self.z) == (o.x, o.y, o.z) ---------
--- def __lt__(self, o): ---------------------------------------------
------- return (self.x, self.y, self.z) < (o.x, o.y, o.z) ----------
 ----------------------------------------------------------------------
class Face: ------------------------------------------------------------
--- __slots__ = ['p', 'q'] -------------------------------------------
--- def __init__(self, p, q): ----------------------------------------
------- self.p = p # Tuple of 3 point indices ----------------------
------- self.q = q # Normal vector (P3 object) ----------------------
 ----------------------------------------------------------------------
def hull_3D(pts): ------------------------------------------------------
--- n = len(pts) -----------------------------------------------------
--- faces = [] -------------------------------------------------------
--- dead = [[True] * n for _ in range(n)] ----------------------------
--- def add_face(a, b, c): -------------------------------------------
------- q = (pts[b] - pts[a]).cross(pts[c] - pts[a]) ---------------
------- faces.append(Face((a, b, c), q)) ---------------------------
------- dead[a][b] = dead[b][c] = dead[c][a] = False ---------------
--- # Initialize with a degenerate tetrahedron ---------------------
--- add_face(0, 1, 2) ------------------------------------------------
--- add_face(0, 2, 1) ------------------------------------------------
--- for i in range(3, n): --------------------------------------------
------- f_inv = [] -------------------------------------------------
------- for f in faces: --------------------------------------------
----------- # If face is visible from the point ---------------------
----------- if (pts[i] - pts[f.p[0]]).dot(f.q) > 0: ----------------
--------------- for j in range(3): --------------------------------
------------------ dead[f.p[j]][f.p[(j+1)%3]] = True ---------------
----------- else: ----------------------------------------------------
--------------- f_inv.append(f) -----------------------------------
------- faces = [] -------------------------------------------------
------- for f in f_inv: --------------------------------------------
----------- for j in range(3): ---------------------------------------
--------------- a, b = f.p[j], f.p[(j+1)%3] -----------------------
--------------- # If reverse edge is dead but forward is alive -------
--------------- if dead[b][a]: -----------------------------------
------------------ add_face(b, a, i) ---------------------------------
------- faces.extend(f_inv) ----------------------------------------
--- return faces -----------------------------------------------------
```

3.7. Areas.

3.7.1. *Triangle Area.*

```
def area(a, b, c): -----------------------------------------------------
--- s = (a + b + c) / 2 ----------------------------------------------
--- return (s*(s-a)*(s-b)*(s-c)) ** 0.5 ------------------------------
```

3.7.2. *Polygon Area.*

```
def area(p, n): --------------------------------------------------------
--- a = 0 ------------------------------------------------------------
--- for i in range(n): -----------------------------------------------
------- j = (i - 1) % n --------------------------------------------
------- a += cross(p[i], p[j]) -------------------------------------
--- return abs(a) / 2 ------------------------------------------------
```

3.8. Dot Cross.

```
def dot(a, b):
    return a.x * b.x + a.y * b.y # + a.z * b.z
def cross(a, b):
    return a.x * b.y - a.y * b.x
def cross(a, b, c):
    return cross(a, b) + cross(b, c) + cross(c, a)
def cross3D(a, b):
    return point3(a.x*b.y - a.y*b.x, a.y*b.z - a.z*b.y, a.z*b.x - a.x*b.z)
```

3.9. Line Sweep Closest Pair.

```
import math

def dist_sq(p1, p2):
    return (p1[0] - p2[0])**2 + (p1[1] - p2[1])**2

def closest_pair(points):
    points.sort()

    def solve(pts):
        n = len(pts)

        if n <= 3:
            return min(dist_sq(pts[i], pts[j])
                       for i in range(n)
                       for j in range(i + 1, n)) if n > 1 else float('inf')

        mid = n // 2
        mid_x = pts[mid][0]

        left = pts[:mid]
        right = pts[mid:]

        d = min(solve(left), solve(right))

        strip = [p for p in pts if (p[0] - mid_x) ** 2 < d]
        strip.sort(key=lambda p: p[1])

        for i in range(len(strip)):
            for j in range(i + 1, min(i + 7, len(strip))):
                d = min(d, dist_sq(strip[i], strip[j]))

        return d

    return solve(points)
```

3.10. Point Projection.

```
EPS = 1e-9

def dot(a, b):
    return sum(x * y for x, y in zip(a, b))

def scale(v, k):
    return [k * x for x in v]

def add(a, b):
    return [x + y for x, y in zip(a, b)]

def sub(a, b):
    return [x - y for x, y in zip(a, b)]

def norm(v):
    return dot(v, v)

def proj(p, v):
    v_norm = norm(v)
    if v_norm < EPS:
        return [0] * len(p)
    return scale(v, dot(p, v) / v_norm)

def proj_line(p, a, b):
    ab = sub(b, a)
    ab_norm = norm(ab)
    if ab_norm < EPS:
        return a[:]

    t = dot(sub(p, a), ab) / ab_norm
    return add(a, scale(ab, t))

def proj_seg(p, a, b):
    ab = sub(b, a)
    ab_norm = norm(ab)
    if ab_norm < EPS:
        return a[:]

    t = dot(sub(p, a), ab) / ab_norm
    t = max(0.0, min(1.0, t))

    return add(a, scale(ab, t))

def proj_plane(p, a, b, c, d):
    n = [a, b, c]
    n_norm = norm(n)
    if n_norm < EPS:
        return p[:]

    dist = (dot(p, n) + d) / n_norm
    return sub(p, scale(n, dist))
```

3.11. Great Circle Distance.

```
import math

PI = math.pi

def clamp(x, lo, hi):
    return max(lo, min(hi, x))

# ========================================================
# version 1: lat/long (spherical law of cosines)
# ========================================================
def great_circle_dist_latlong(lat1, lon1, lat2, lon2, R=6371.0):
    lat1 *= PI / 180
    lon1 *= PI / 180
    lat2 *= PI / 180
    lon2 *= PI / 180

    val = (
        math.sin(lat1) * math.sin(lat2)
        + math.cos(lat1) * math.cos(lat2) * math.cos(abs(lon1 - lon2))
    )

    val = clamp(val, -1.0, 1.0)  # IMPORTANT FIX

    return R * math.acos(val)

def dot(a, b):
    return sum(x * y for x, y in zip(a, b))

def cross(a, b):
    return [
        a[1] * b[2] - a[2] * b[1],
        a[2] * b[0] - a[0] * b[2],
        a[0] * b[1] - a[1] * b[0],
    ]

def norm(v):
    return math.sqrt(dot(v, v))

# ========================================================
# version 2: using 3D vectors (angle between vectors)
# ========================================================
def great_circle_dist_vec(a, b, R=1.0):
    # assumes a and b are unit vectors on sphere
    cross_prod = cross(a, b)
    return R * math.atan2(norm(cross_prod), dot(a, b))

def to_vec(lat, lon):
    lat *= PI / 180
    lon *= PI / 180

    return [
        math.cos(lat) * math.cos(lon),
        math.cos(lat) * math.sin(lon),
        math.sin(lat),
    ]
```

3.12. Point In Polygon.

```
import math

EPS = 1e-9

def dist(a, b):
    return math.hypot(a[0] - b[0], a[1] - b[1])

def in_polygon(q, p):
    n = len(p)
    inside = False

    j = n - 1
    for i in range(n):
        xi, yi = p[i]
        xj, yj = p[j]
        xq, yq = q

        intersect = ((yi > yq) != (yj > yq)) and (
            xq < (xj - xi) * (yq - yi) / (yj - yi + EPS) + xi
        )

        if intersect:
            inside = not inside

        j = i

    return inside

def on_polygon(q, p):
    n = len(p)

    for i in range(n):
        j = (i + 1) % n

        a = p[i]
        b = p[j]

        if abs(dist(a, q) + dist(q, b) - dist(a, b)) < EPS:
            return True

    return False
```

3.13. Cut Polygon By Line.

```
EPS = 1e-9

def cross(a, b, c):
    return (b[0] - a[0]) * (c[1] - a[1]) - (b[1] - a[1]) * (c[0] - a[0])
```

```
def line_inter(p1, p2, a, b):
    x1, y1 = p1
    x2, y2 = p2
    x3, y3 = a
    x4, y4 = b

    den = (x1 - x2) * (y3 - y4) - (y1 - y2) * (x3 - x4)
    if abs(den) < EPS:
        return p1

    px = (
        (x1 * y2 - y1 * x2) * (x3 - x4)
        - (x1 - x2) * (x3 * y4 - y3 * x4)
    ) / den

    py = (
        (x1 * y2 - y1 * x2) * (y3 - y4)
        - (y1 - y2) * (x3 * y4 - y3 * x4)
    ) / den

    return (px, py)

def cut(poly, a, b):
    n = len(poly)
    res = []

    for i in range(n):
        j = (i - 1 + n) % n

        c1 = cross(a, b, poly[j])
        c2 = cross(a, b, poly[i])

        if c1 >= -EPS:
            res.append(poly[j])

        if c1 * c2 < -EPS:
            inter = line_inter(poly[j], poly[i], a, b)
            res.append(inter)

    return res
```

3.14. Triangle Centers.

```
import math

EPS = 1e-9

def add(a, b):
    return (a[0] + b[0], a[1] + b[1])
def sub(a, b):
    return (a[0] - b[0], a[1] - b[1])
def scale(a, k):
    return (a[0] * k, a[1] * k)
def norm2(a, b):
    return (a[0] - b[0]) ** 2 + (a[1] - b[1]) ** 2
def bary(A, B, C, a, b, c):
    s = a + b + c
    if abs(s) < EPS:
        return (0.0, 0.0)

    return (
        (A[0] * a + B[0] * b + C[0] * c) / s,
        (A[1] * a + B[1] * b + C[1] * c) / s,
    )
def dist(a, b):
    return math.sqrt(norm2(a, b))
def trilinear(A, B, C, a, b, c):
    return bary(
        A,
        B,
        C,
        dist(B, C) * a,
        dist(C, A) * b,
        dist(A, B) * c,
    )
def centroid(A, B, C):
    return bary(A, B, C, 1, 1, 1)
def circumcenter(A, B, C):
    a = dist(B, C)
    b = dist(C, A)
    c = dist(A, B)

    return bary(
        A,
        B,
        C,
        a * (b + c - a),
        b * (c + a - b),
        c * (a + b - c),
    )
def orthocenter(A, B, C):
    def foot(P, Q, R):
        q = sub(Q, R)
        t = dot(sub(P, R), q) / (dot(q, q) + EPS)
        return add(R, scale(q, t))
    def dot(a, b):
        return a[0] * b[0] + a[1] * b[1]

    H1 = foot(A, B, C)
    H2 = foot(B, A, C)

    return bary(A, B, C, 1, 1, 1)
def incenter(A, B, C):
    return bary(
        A,
        B,
        C,
        dist(B, C),
        dist(C, A),
        dist(A, B),
    )
def inradius(a, b, c):
    s = (a + b + c) / 2
    area = math.sqrt(max(s * (s - a) * (s - b) * (s - c), 0))
    return area / s if s > EPS else 0
def excenter(A, B, C):
    a = dist(B, C)
    b = dist(C, A)
    c = dist(A, B)
    return bary(A, B, C, -a, b, c)
def symmedian(A, B, C):
    return bary(
        A,
        B,
        C,
        norm2(B, C),
        norm2(C, A),
        norm2(A, B),
    )
```

3.15. Convex Polygon Intersection.

```
EPS = 1e-9

def sub(a, b):
    return (a[0] - b[0], a[1] - b[1])
def add(a, b):
    return (a[0] + b[0], a[1] + b[1])
def cross(a, b):
    return a[0] * b[1] - a[1] * b[0]
def dot(a, b):
    return a[0] * b[0] + a[1] * b[1]
def orient(a, b, c):
    return cross(sub(b, a), sub(c, a))
def seg_inter(a, b, c, d):
    def on_seg(p, a, b):
        return (
            abs(orient(a, b, p)) < EPS
            and min(a[0], b[0]) - EPS <= p[0] <= max(a[0], b[0]) + EPS
            and min(a[1], b[1]) - EPS <= p[1] <= max(a[1], b[1]) + EPS
        )
    def line_inter(a, b, c, d):
        A1 = b[1] - a[1]
        B1 = a[0] - b[0]
        C1 = A1 * a[0] + B1 * a[1]
        A2 = d[1] - c[1]
        B2 = c[0] - d[0]
        C2 = A2 * c[0] + B2 * c[1]
        det = A1 * B2 - A2 * B1
        if abs(det) < EPS:
            return None
        x = (B2 * C1 - B1 * C2) / det
        y = (A1 * C2 - A2 * C1) / det
        return (x, y)
    p = line_inter(a, b, c, d)
    if p is None:
        return None
    if (
        on_seg(p, a, b)
        and on_seg(p, c, d)
    ):
        return p
    return None
def in_poly(p, poly):
    inside = False
    n = len(poly)
    for i in range(n):
        j = (i - 1) % n
        a, b = poly[i], poly[j]
        if ((a[1] > p[1]) != (b[1] > p[1])):
            x = (b[0] - a[0]) * (p[1] - a[1]) / (b[1] - a[1] + EPS) +
↪   a[0]
            if p[0] < x:
                inside = not inside
    return inside
def on_poly(p, poly):
    n = len(poly)
    for i in range(n):
        j = (i - 1) % n
        if abs(orient(poly[j], poly[i], p)) < EPS:
            return True
    return False
def convex_hull(points):
    points = sorted(points)
    def build(h):
        for p in points:
            while len(h) >= 2 and orient(h[-2], h[-1], p) <= EPS:
                h.pop()
            h.append(p)
        return h
    lower = build([])
    upper = build([])
    return lower[:-1] + upper[:-1]
def convex_polygon_inter(a, b):
    ans = []
    for p in a:
        if in_poly(p, b) or on_poly(p, b):
            ans.append(p)
    for p in b:
        if in_poly(p, a) or on_poly(p, a):
            ans.append(p)
    for i in range(len(a)):
        I = (i - 1) % len(a)
        for j in range(len(b)):
            J = (j - 1) % len(b)
            p = seg_inter(a[I], a[i], b[J], b[j])
```

```
------------ if p is not None:
--------------- ans.append(p)
--- if not ans:
------- return []
--- return convex_hull(ans)
```

3.16. Pick Theorem Lattice.

```
import math

def area(poly):
--- n = len(poly)
--- s = 0
--- for i in range(n):
------- j = (i - 1) % n
------- s += poly[j][0] * poly[i][1] - poly[i][0] * poly[j][1]
--- return abs(s) // 2
def boundary(poly):
--- n = len(poly)
--- ans = 0
--- for i in range(n):
------- j = (i - 1) % n
------- dx = poly[i][0] - poly[j][0]
------- dy = poly[i][1] - poly[j][1]
------- ans += math.gcd(abs(dx), abs(dy))
--- return ans
def interior(poly):
--- A = area(poly)
--- B = boundary(poly)
--- return A - B // 2 + 1
```

3.17. Minimum Enclosing Circle.

```
import math

EPS = 1e-9
def dist(a, b):
--- return math.hypot(a[0] - b[0], a[1] - b[1])
def mid(a, b):
--- return ((a[0] + b[0]) / 2, (a[1] + b[1]) / 2)
def cross(a, b, c):
--- return (b[0] - a[0]) * (c[1] - a[1]) - (b[1] - a[1]) * (c[0] -
↪  a[0])
def circumcenter(a, b, c):
--- ax, ay = a
--- bx, by = b
--- cx, cy = c
--- d = 2 * (ax * (by - cy) + bx * (cy - ay) + cx * (ay - by))
--- if abs(d) < EPS:
------- return mid(a, b)
--- ux = (
------- (ax * ax + ay * ay) * (by - cy)
------- + (bx * bx + by * by) * (cy - ay)
------- + (cx * cx + cy * cy) * (ay - by)
--- ) / d
--- uy = (
------- (ax * ax + ay * ay) * (cx - bx)
------- + (bx * bx + by * by) * (ax - cx)
------- + (cx * cx + cy * cy) * (bx - ax)
--- ) / d
--- return (ux, uy)
def bounding_ball(p):
--- random.shuffle(p)
--- center = p[0]
--- radius = 0.0
--- n = len(p)
--- for i in range(n):
------- if dist(center, p[i]) > radius + EPS:
----------- center = p[i]
----------- radius = 0.0
----------- for j in range(i):
--------------- if dist(center, p[j]) > radius + EPS:
------------------- center = mid(p[i], p[j])
------------------- radius = dist(center, p[i])
------------------- for k in range(j):
----------------------- if dist(center, p[k]) > radius + EPS:
--------------------------- center = circumcenter(p[i], p[j], p[k])
--------------------------- radius = dist(center, p[i])
--- return center, radius
def verify(p, center, r):
--- for x in p:
------- if dist(x, center) > r + 1e-6:
----------- return False
--- return True
```

3.18. Shamos Algorithm.

```
import math

EPS = 1e-9
def sub(a, b):
--- return (a[0] - b[0], a[1] - b[1])
def cross(a, b):
--- return a[0] * b[1] - a[1] * b[0]
def dist_pt_line(p, a, b):
--- ab = sub(b, a)
--- ap = sub(p, a)
--- return abs(cross(ab, ap)) / (math.hypot(ab[0], ab[1]) + EPS)
def convex_hull(points):
--- points = sorted(points)
--- def build():
------- hull = []
------- for p in points:
----------- while len(hull) >= 2 and cross(sub(hull[-1], hull[-2]),
↪  sub(p, hull[-1])) <= EPS:
--------------- hull.pop()
----------- hull.append(p)
------- return hull
--- lower = build()
--- upper = build()
--- return lower[:-1] + upper[:-1]
def shamos(p):
--- h = convex_hull(p)
--- k = len(h)

--- if k <= 2:
------- return 0.0

--- d = float("inf")
--- j = 1

--- for i in range(k):
------- ni = (i + 1) % k

------- a = h[i]
------- b = h[ni]

------- while True:
----------- nj = (j + 1) % k
----------- if dist_pt_line(h[nj], a, b) > dist_pt_line(h[j], a, b):
--------------- j = nj
----------- else:
--------------- break
------- d = min(d, dist_pt_line(h[j], a, b))
--- return d
```

3.19. Kd Tree.

```
import heapq

EPS = 1e-9
def dist2(a, b):
--- return (a[0] - b[0]) ** 2 + (a[1] - b[1]) ** 2
class KDTree:
--- def __init__(self, points):
------- self.p = points[:]
------- self.n = len(points)
------- self.build(0, self.n, 0)
--- def build(self, L, R, dvx):
------- if R - L <= 1:
----------- return
------- mid = (L + R) // 2
------- if dvx:
----------- self.p[L:R] = sorted(self.p[L:R], key=lambda x: x[0])
------- else:
----------- self.p[L:R] = sorted(self.p[L:R], key=lambda x: x[1])
------- self.build(L, mid, not dvx)
------- self.build(mid + 1, R, not dvx)
--- def knn(self, x, y, k=1, r=-1):
------- self.qx = x
------- self.qy = y
------- self.k = k
------- self.prune = float("inf") if r < 0 else r * r
------- self.heap = []
------- self._dfs(0, self.n, False)
------- res = [p for _, p in sorted(self.heap, reverse=True)]
------- return res
--- def _dfs(self, L, R, dvx):
------- if R - L <= 0:
----------- return
------- mid = (L + R) // 2
------- p = self.p[mid]
------- dx = self.qx - p[0]
------- dy = self.qy - p[1]
------- D = dx * dx + dy * dy
------- if D <= self.prune:
----------- heapq.heappush(self.heap, (-D, p))
----------- if len(self.heap) > self.k:
--------------- heapq.heappop(self.heap)
------- delta = dx if dvx else dy
------- nL, nR = L, mid
------- fL, fR = mid + 1, R
------- if delta > 0:
----------- nL, fL = fL, nL
----------- nR, fR = fR, nR
------- self._dfs(nL, nR, not dvx)
------- if len(self.heap) < self.k or delta * delta <= self.prune:
----------- self._dfs(fL, fR, not dvx)
```

4. Graphs

4.1. Max Flow.

4.1.1. *Dinic.*

```
INF = float('inf')
class Dinic:
--- def __init__(self, n, s, t):
------- self.n, self.s, self.t = n, s, t
------- self.edges = []
------- self.adj = [[] for _ in range(n)]

--- def add_edge(self, u, v, c, bi=False):
------- self.adj[u].append(len(self.edges))
------- self.edges.append([v, c, 0])
------- self.adj[v].append(len(self.edges))
------- self.edges.append([u, c if bi else 0, 0])

--- def bfs(self):
------- self.dist = [-1] * self.n
------- self.dist[self.s] = 0
------- q = deque([self.s])
------- while q:
----------- u = q.popleft()
----------- for i in self.adj[u]:
--------------- v, c, f = self.edges[i]
```

```
---------------- if self.dist[v] < 0 and c > f:
-------------------- self.dist[v] = self.dist[u] + 1
-------------------- q.append(v)
------- return self.dist[self.t] != -1
---
--- def dfs(self, u, pushed):
------- if pushed == 0 or u == self.t: return pushed
------- while self.ptr[u] < len(self.adj[u]):
----------- i = self.adj[u][self.ptr[u]]
----------- v, c, f = self.edges[i]
----------- if self.dist[u]+1 == self.dist[v] and c > f:
---------------- tr = self.dfs(v, min(pushed, c - f))
---------------- if tr > 0:
-------------------- self.edges[i][2] += tr
-------------------- self.edges[i^1][2] -= tr
-------------------- return tr
----------- self.ptr[u] += 1
------- return 0
---
--- def calc_max_flow(self):
------- flow = 0
------- while self.bfs():
----------- self.ptr = [0] * self.n
----------- while True:
---------------- pushed = self.dfs(self.s, INF)
---------------- if not pushed:
-------------------- break
---------------- flow += pushed
------- return flow
```

4.1.2. *Edmonds Karp.*

```
INF = float('inf')
class EdmondsKarp:
--- def __init__(self, n, s, t):
------- self.n, self.s, self.t = n, s, t
------- self.adj = [[] for _ in range(n)]
------- self.res = [[0] * n for _ in range(n)]
---
--- def add_edge(self, u, v, w):
------- self.adj[u].append(v)
------- self.adj[v].append(u)
------- self.res[u][v] += w
---
--- def bfs(self):
------- self.par = [-1] * self.n
------- self.par[self.s] = self.s
------- q = deque([self.s])
------- while q:
----------- u = q.popleft()
----------- for v in self.adj[u]:
---------------- if self.res[u][v] > 0 and self.par[v] == -1:
-------------------- self.par[v] = u
-------------------- if v == self.t:
------------------------ return True
-------------------- q.append(v)
------- return False
---
--- def calc_max_flow(self):
------- ans = 0
------- while self.bfs():
----------- flow = INF
----------- u = self.t
----------- while u != self.s:
---------------- p = self.par[u]
---------------- flow = min(flow, self.res[p][u])
---------------- u = p
----------- u = self.t
----------- while u != self.s:
---------------- p = self.par[u]
---------------- self.res[p][u] -= flow
---------------- self.res[u][p] += flow
---------------- u = p
----------- ans += flow
------- return ans
```

4.1.3. *Bipartite Matching.*

```
from collections import deque
---
INF = float('inf')
class BipartiteGraph:
--- def __init__(self, n, m):
------- self.n = n
------- self.m = m
------- self.adj = [[] for _ in range(n)]
------- self.L = [-1] * n
------- self.R = [-1] * m
------- self.dist = [INF] * (n + 1)
---
--- def add_edge(self, u, v):
------- self.adj[u].append(v)
---
--- def bfs(self):
------- queue = deque()
------- for v in range(self.n):
----------- if self.L[v] == -1:
---------------- self.dist[v] = 0
---------------- queue.append(v)
----------- else:
---------------- self.dist[v] = INF
---
------- self.dist[self.n] = INF
---
------- while queue:
----------- v = queue.popleft()
----------- if self.dist[v] < self.dist[self.n]:
---------------- for u in self.adj[v]:
-------------------- next_v = self.R[u]
-------------------- dist_next = self.dist[next_v] if next_v != -1 else
↪  self.dist[self.n]
---
-------------------- if (next_v == -1 and self.dist[self.n] == INF) or \
------------------------ (next_v != -1 and self.dist[next_v] ==
↪  INF):
---
---------------------- target = next_v if next_v != -1 else self.n --
---------------------- self.dist[target] = self.dist[v] + 1 ---------
---------------------- if target != self.n:
------------------------ queue.append(target)
---
------- return self.dist[self.n] != INF
---
--- def dfs(self, v):
------- if v != -1:
----------- for u in self.adj[v]:
---------------- next_v = self.R[u]
---------------- dist_next = self.dist[next_v] if next_v != -1 else
↪  self.dist[self.n]
---
---------------- if dist_next == self.dist[v] + 1:
-------------------- if self.dfs(next_v):
---------------------- self.R[u] = v
---------------------- self.L[v] = u
---------------------- return True
----------- self.dist[v] = INF
----------- return False
------- return True
---
--- def maximum_matching(self):
------- matching = 0
------- for u in range(self.n):
----------- for v in self.adj[u]:
---------------- if self.R[v] == -1:
-------------------- self.L[u] = v
-------------------- self.R[v] = u
-------------------- matching += 1
-------------------- break
---
------- while self.bfs():
----------- for i in range(self.n):
---------------- if self.L[i] == -1 and self.dfs(i):
-------------------- matching += 1
------- return matching
# Input Format: g = BipartiteGraph(2, 2)
#               g.add_edge(0, 0)
#               g.add_edge(0, 1)
#               g.add_edge(1, 0)
#               print(f"Max Matching: {g.maximum_matching()}")
```

4.2. Shortest Path.

4.2.1. *Dijkstra.*

```
import heapq
INF = float('inf')
def dijkstra(n, adj, s):
--- dist = [INF] * n
--- dist[s] = 0
--- pq = [(0, s)]
---
--- while pq:
------- d, u = heapq.heappop(pq)
------- if dist[u] < d:
----------- continue
------- for v, w in adj[u]:
----------- if dist[v] > d + w:
---------------- dist[v] = d + w
---------------- heapq.heappush(pq, (dist[v], v))
--- return dist
```

4.2.2. *Bellman Ford.*

```
INF = float('inf')
def bellman_ford(n, adj, s):
--- dist = [INF] * n
--- dist[s] = 0
--- for _ in range(n - 1):
------- for u in range(n):
----------- if dist[u] == INF: continue
----------- for v, w in adj[u]:
---------------- if dist[u] + w < dist[v]:
-------------------- dist[v] = dist[u] + w
--- for u in range(n):
------- if dist[u] == INF: continue
------- for v, w in adj[u]:
----------- if dist[u] + w < dist[v]:
---------------- return dist, True
--- return dist, False
```

4.3. Minimum Spanning Tree.

4.3.1. *Kruskal Min Spanning Tree.*

```
import heapq
---
class UnionFind:
--- def __init__(self, n):
------- self.parent = list(range(n))
------- self.rank = [0] * n
---
--- def find(self, x):
------- while self.parent[x] != x:
----------- self.parent[x] = self.parent[self.parent[x]]
----------- x = self.parent[x]
------- return x
```

```
    def unite(self, a, b):
        a = self.find(a)
        b = self.find(b)

        if a == b:
            return False

        if self.rank[a] < self.rank[b]:
            self.parent[a] = b
        else:
            self.parent[b] = a
            if self.rank[a] == self.rank[b]:
                self.rank[a] += 1

        return True

def kruskal(n, edges):
    edges = edges[:]
    heapq.heapify(edges)

    uf = UnionFind(n)
    total_weight = 0
    mst = []

    while edges:
        w, u, v = heapq.heappop(edges)
        if uf.unite(u, v):
            mst.append((u, v, w))
            total_weight += w

            if len(mst) == n - 1:
                break

    return mst, total_weight
```

4.3.2. *Prim Min Spanning Tree.*

```
import heapq

def prim(n, adj, start=0):
    visited = [False] * n
    pq = [(0, start, -1)]

    total_weight = 0
    mst = []

    while pq:
        w, u, parent = heapq.heappop(pq)

        if visited[u]:
            continue

        visited[u] = True

        if parent != -1:
            mst.append((parent, u, w))
            total_weight += w

        for v, weight in adj[u]:
            if not visited[v]:
                heapq.heappush(pq, (weight, v, u))

    return mst, total_weight
```

4.4. **Least Common Ancestor.**

4.4.1. *Binary Lifting Lca.*

```
import math

class BinaryLifting:
    def __init__(self, adj):
        self.n = len(adj)
        self.LOG = math.ceil(math.log2(self.n)) + 1
        self.adj = adj

        self.dep = [0] * self.n
        self.par = [[-1] * self.LOG for _ in range(self.n)]

        self.build(0)

    def dfs(self, u, p):
        self.par[u][0] = p
        for v in self.adj[u]:
            if v != p:
                self.dep[v] = self.dep[u] + 1
                self.dfs(v, u)

    def build(self, root):
        self.dfs(root, root)

        for k in range(1, self.LOG):
            for u in range(self.n):
                self.par[u][k] = self.par[self.par[u][k-1]][k-1]

    def lift(self, u, k):
        for i in range(self.LOG):
            if k & (1 << i):
                u = self.par[u][i]
        return u

    def lca(self, u, v):
        if self.dep[u] < self.dep[v]:
            u, v = v, u

        u = self.lift(u, self.dep[u] - self.dep[v])

        if u == v:
            return u

        for k in reversed(range(self.LOG)):
            if self.par[u][k] != self.par[v][k]:
                u = self.par[u][k]
                v = self.par[v][k]

        return self.par[u][0]
```

4.4.2. *Euler Lca.*

```
class EulerLCA:
    def __init__(self, adj):
        self.n = len(adj)
        self.adj = adj

        self.euler = []
        self.first = [-1] * self.n
        self.depth = [0] * self.n

        self.build(0)

    def dfs(self, u, p, d):
        self.first[u] = len(self.euler)
        self.depth[u] = d
        self.euler.append(u)

        for v in self.adj[u]:
            if v != p:
                self.dfs(v, u, d + 1)
                self.euler.append(u)

    def build(self, root):
        self.dfs(root, root, 0)

        m = len(self.euler)
        LOG = m.bit_length()

        self.st = [[0] * m for _ in range(LOG)]
        self.log = [0] * (m + 1)

        for i in range(2, m + 1):
            self.log[i] = self.log[i // 2] + 1

        for i in range(m):
            self.st[0][i] = self.euler[i]

        for k in range(1, LOG):
            for i in range(m - (1 << k) + 1):
                a = self.st[k-1][i]
                b = self.st[k-1][i + (1 << (k-1))]
                self.st[k][i] = a if self.depth[a] < self.depth[b] else
↪   b

    def lca(self, u, v):
        l, r = self.first[u], self.first[v]
        if l > r:
            l, r = r, l

        k = self.log[r - l + 1]
        a = self.st[k][l]
        b = self.st[k][r - (1 << k) + 1]

        return a if self.depth[a] < self.depth[b] else b
```

4.4.3. *Tarjan Lca.*

```
class DSU:
    def __init__(self, n):
        self.p = list(range(n))

    def find(self, x):
        if self.p[x] != x:
            self.p[x] = self.find(self.p[x])
        return self.p[x]

    def unite(self, a, b):
        self.p[self.find(b)] = self.find(a)


class Tarjan:
    def __init__(self, adj, queries):
        self.adj = adj
        self.queries = queries
        self.n = len(adj)

        self.dsu = DSU(self.n)
        self.vis = [False] * self.n
        self.anc = [0] * self.n
        self.res = {}

    def dfs(self, u, p):
        self.anc[u] = u

        for v in self.adj[u]:
            if v == p:
                continue
            self.dfs(v, u)
            self.dsu.unite(u, v)
            self.anc[self.dsu.find(u)] = u

        self.vis[u] = True

        for v, idx in self.queries[u]:
            if self.vis[v]:
                self.res[idx] = self.anc[self.dsu.find(v)]
```

4.5. Minimum Mean Weight Cycle.

```
def min_mean_cycle(adj):
    n = len(adj)
    INF = float('inf')

    arr = [[INF] * n for _ in range(n + 1)]

    for i in range(n):
        arr[0][i] = 0

    for k in range(1, n + 1):
        for u in range(n):
            if arr[k - 1][u] == INF:
                continue
            for v, w in adj[u]:
                arr[k][v] = min(arr[k][v], arr[k - 1][u] + w)

    result = INF

    for v in range(n):
        if arr[n][v] == INF:
            continue

        max_avg = -INF
        for k in range(n):
            if arr[k][v] == INF:
                continue
            avg = (arr[n][v] - arr[k][v]) / (n - k)
            max_avg = max(max_avg, avg)

        result = min(result, max_avg)

    return result
```

4.6. Strongly Connected Components.

```
class KosarajuGraph:
    def __init__(self, n):
        self.n = n
        self.graph = [[] for _ in range(n)]
        self.rev_graph = [[] for _ in range(n)]
        self.vis = [0] * n
        self.sccs = []

    def add_edge(self, u, v):
        self.graph[u].append(v)
        self.rev_graph[v].append(u)

    def dfs(self, u, adj, container):
        self.vis[u] = 1
        for v in adj[u]:
            if not self.vis[v]:
                self.dfs(v, adj, container)
        container.append(u)

    def kosaraju(self):
        topo = []

        # First pass
        self.vis = [0] * self.n
        for i in range(self.n):
            if not self.vis[i]:
                self.dfs(i, self.graph, topo)

        # Second pass
        self.vis = [0] * self.n
        self.sccs = []

        for u in reversed(topo):
            if not self.vis[u]:
                comp = []
                self.dfs(u, self.rev_graph, comp)
                self.sccs.append(comp)

        return self.sccs
```

4.7. Heavy Light Decomposition.

```
class SegmentTree:
    def __init__(self, n):
        self.n = n
        self.tree = [0] * (4 * n)

    # Example: point update
    def update(self, idx, val, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if l == r:
            self.tree[node] = vala
            return
        mid = (l + r) // 2
        if idx <= mid:
            self.update(idx, val, node * 2, l, mid)
        else:
            self.update(idx, val, node * 2 + 1, mid + 1, r)
        self.tree[node] = self.tree[node * 2] + self.tree[node * 2 + 1]

    # Example: range query
    def query(self, ql, qr, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if qr < l or r < ql:
            return 0
        if ql <= l and r <= qr:
            return self.tree[node]
        mid = (l + r) // 2
        return self.query(ql, qr, node * 2, l, mid) + \
               self.query(ql, qr, node * 2 + 1, mid + 1, r)

class HeavyLightTree:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.par = [-1] * n
        self.heavy = [-1] * n
        self.dep = [0] * n
        self.path_root = [0] * n
        self.pos = [0] * n

        self.segment_tree = SegmentTree(n)
        self.current_pos = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.adj[v].append(u)

    def dfs(self, u):
        size = 1
        max_subtree_size = 0

        for v in self.adj[u]:
            if v != self.par[u]:
                self.par[v] = u
                self.dep[v] = self.dep[u] + 1
                subtree_size = self.dfs(v)

                if subtree_size > max_subtree_size:
                    max_subtree_size = subtree_size
                    self.heavy[u] = v

                size += subtree_size

        return size

    def decompose(self, u, root):
        self.path_root[u] = root
        self.pos[u] = self.current_pos
        self.current_pos += 1

        if self.heavy[u] != -1:
            self.decompose(self.heavy[u], root)

        for v in self.adj[u]:
            if v != self.par[u] and v != self.heavy[u]:
                self.decompose(v, v)

    def build(self, root=0):
        self.par[root] = root
        self.dep[root] = 0
        self.dfs(root)
        self.decompose(root, root)

    # Query path u -> v
    def query(self, u, v):
        res = 0
        while self.path_root[u] != self.path_root[v]:
            if self.dep[self.path_root[u]] <
↪  self.dep[self.path_root[v]]:
                u, v = v, u
            start = self.path_root[u]
            res += self.segment_tree.query(self.pos[start],
↪  self.pos[u])
            u = self.par[start]

        # Same path
        if self.dep[u] > self.dep[v]:
            u, v = v, u
        res += self.segment_tree.query(self.pos[u], self.pos[v])
        return res

    # Update node value
    def update(self, u, value):
        self.segment_tree.update(self.pos[u], value)
```

4.8. Euler Path/Cycle.

4.8.1. *Directed Euler.*

```
from collections import defaultdict

class DirectedEuler:
    def __init__(self, n):
        self.n = n
        self.adj = defaultdict(list)
        self.indeg = [0]*n
        self.outdeg = [0]*n
        self.edge_count = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.outdeg[u] += 1
        self.indeg[v] += 1
        self.edge_count += 1

    def find_start_node(self):
        start = -1
        end = -1
        for i in range(self.n):
            if self.outdeg[i] - self.indeg[i] == 1:
                if start != -1:
                    return -1
                start = i
            elif self.indeg[i] - self.outdeg[i] == 1:
                if end != -1:
                    return -1
```

```
---------------- end = i
------------ elif self.indeg[i] != self.outdeg[i]:
---------------- return -1

-------- if start == -1:
------------ for i in range(self.n):
---------------- if self.outdeg[i] > 0:
-------------------- return i
-------- return start

--- def get_euler_path(self):
-------- start = self.find_start_node()
-------- if start == -1:
------------ return None

-------- stack = [start]
-------- path = []

-------- local_adj = {u: self.adj[u][:] for u in self.adj}

-------- while stack:
------------ u = stack[-1]
------------ if u in local_adj and local_adj[u]:
---------------- v = local_adj[u].pop()
---------------- stack.append(v)
------------ else:
---------------- path.append(stack.pop())

-------- if len(path) != self.edge_count + 1:
------------ return None

-------- return path[::-1]

--- class UndirectedEuler:
-------- def __init__(self, n):
------------ self.n = n
------------ self.adj = defaultdict(list)
------------ self.edge_count = 0

-------- def add_edge(self, u, v):
------------ self.adj[u].append(v)
------------ self.adj[v].append(u)
------------ self.edge_count += 1

-------- def is_valid(self):
------------ odd = 0
------------ for i in range(self.n):
---------------- if len(self.adj[i]) % 2 != 0:
-------------------- odd += 1
------------ return odd == 0 or odd == 2

-------- def get_euler_path(self):
------------ if not self.is_valid():
---------------- return None

------------ start = 0
------------ for i in range(self.n):
---------------- if len(self.adj[i]) % 2 == 1:
-------------------- start = i
-------------------- break
---------------- if len(self.adj[i]) > 0:
-------------------- start = i

------------ stack = [start]
------------ path = []

------------ local_adj = {u: self.adj[u][:] for u in self.adj}

------------ while stack:
---------------- u = stack[-1]
---------------- if u in local_adj and local_adj[u]:
-------------------- v = local_adj[u].pop()
-------------------- local_adj[v].remove(u)
-------------------- stack.append(v)
---------------- else:
-------------------- path.append(stack.pop())

------------ if len(path) != self.edge_count + 1:
---------------- return None

------------ return path[::-1]
```

4.8.2. *Undirected Euler.*

```
from collections import defaultdict

class UndirectedEuler:
--- def __init__(self, n):
-------- self.n = n
-------- self.adj = defaultdict(list)
-------- self.edge_count = 0

--- def add_edge(self, u, v):
-------- self.adj[u].append(v)
-------- self.adj[v].append(u)
-------- self.edge_count += 1

--- def is_valid(self):
-------- odd = 0
-------- for i in range(self.n):
------------ if len(self.adj[i]) % 2 != 0:
---------------- odd += 1
-------- return odd == 0 or odd == 2

--- def get_euler_path(self):
-------- if not self.is_valid():
------------ return None

-------- start = 0
-------- for i in range(self.n):
------------ if len(self.adj[i]) % 2 == 1:
---------------- start = i
---------------- break
------------ if len(self.adj[i]) > 0:
---------------- start = i

-------- stack = [start]
-------- path = []

-------- local_adj = {u: self.adj[u][:] for u in self.adj}

-------- while stack:
------------ u = stack[-1]
------------ if u in local_adj and local_adj[u]:
---------------- v = local_adj[u].pop()
---------------- local_adj[v].remove(u)
---------------- stack.append(v)
------------ else:
---------------- path.append(stack.pop())

-------- if len(path) != self.edge_count + 1:
------------ return None

-------- return path[::-1]
```

4.9. Tree Isomorphism.

```
from collections import deque

def bfs(u, adj, n):
--- vis = [False] * n
--- pre = [-1] * n
--- q = deque()
--- q.append(u)
--- vis[u] = True

--- last = u

--- while q:
------- u = q.popleft()
------- last = u

------- for v in adj[u]:
----------- if not vis[v]:
--------------- vis[v] = True
--------------- pre[v] = u
--------------- q.append(v)

--- return last, pre

def tree_centers(r, adj, n):
--- u, _ = bfs(r, adj, n)
--- v, pre = bfs(u, adj, n)

--- path = []
--- while v != -1:
------- path.append(v)
------- v = pre[v]

--- size = len(path)

--- med = [path[size // 2]]
--- if size % 2 == 0:
------- med.append(path[size // 2 - 1])

--- return med

def rootcode(u, adj, p, d, pr):
--- children = []
--- for v in adj[u]:
------- if v != p:
----------- children.append(rootcode(v, adj, u, (d + 1) % len(pr), pr))

--- children.sort()

--- h = len(children) + 1
--- for x in children:
------- h = h * pr[d] + x

--- return h

def treecode(root, adj, n, pr):
--- c = tree_centers(root, adj, n)

--- if len(c) == 1:
------- return (rootcode(c[0], adj, -1, 0, pr) << 1) | 1
--- else:
------- return (rootcode(c[0], adj, -1, 0, pr) *
--------------- rootcode(c[1], adj, -1, 0, pr)) << 1

def isomorphic(r1, adj1, r2, adj2, n1, n2, pr, rooted=False):
--- if rooted:
------- return rootcode(r1, adj1, -1, 0, pr) == rootcode(r2, adj2, -1,
↪  0, pr)
--- return treecode(r1, adj1, n1, pr) == treecode(r2, adj2, n2, pr)
```

4.10. Centroid Decomposition.

```
import sys
sys.setrecursionlimit(10**7)

INF = 10**9

class CentroidDecomposition:
```

```
--- def __init__(self, n):
------- self.n = n
------- self.adj = [[] for _ in range(n)]

------- self.LG = 20
------- self.jmp = [[-1] * self.LG for _ in range(n)]
------- self.path = [[INF] * self.LG for _ in range(n)]

------- self.sz = [0] * n
------- self.seph = [0] * n
------- self.shortest = [INF] * n
------- self.blocked = [False] * n

--- def add_edge(self, a, b):
------- self.adj[a].append(b)
------- self.adj[b].append(a)

--- def dfs(self, u, p):
------- self.sz[u] = 1
------- for v in self.adj[u]:
----------- if v != p and not self.blocked[v]:
--------------- self.sz[u] += self.dfs(v, u)
------- return self.sz[u]

--- def find_centroid(self, u, p, n):
------- for v in self.adj[u]:
----------- if v != p and not self.blocked[v] and self.sz[v] > n // 2:
--------------- return self.find_centroid(v, u, n)
------- return u

--- def makepaths(self, sep, u, p, depth, h):
------- self.jmp[u][h] = sep
------- self.path[u][h] = depth

------- for v in self.adj[u]:
----------- if v != p and not self.blocked[v]:
--------------- self.makepaths(sep, v, u, depth + 1, h)

--- def separate(self, h=0, u=0):
------- self.dfs(u, -1)
------- c = self.find_centroid(u, -1, self.sz[u])

------- self.seph[c] = h
------- self.makepaths(c, c, -1, 0, h)

------- self.blocked[c] = True

------- for v in self.adj[c]:
----------- if not self.blocked[v]:
--------------- self.separate(h + 1, v)

--- def paint(self, u):
------- for h in range(self.seph[u] + 1):
----------- c = self.jmp[u][h]
----------- self.shortest[c] = min(self.shortest[c], self.path[u][h])

--- def closest(self, u):
------- ans = INF
------- for h in range(self.seph[u] + 1):
----------- c = self.jmp[u][h]
----------- ans = min(ans, self.shortest[c] + self.path[u][h])
------- return ans
```

4.11. Minimum Arborescence.

```
INF = 10**18

class Arborescence:
--- def __init__(self, n):
------- self.n = n
------- self.edges = []

--- def add_edge(self, u, v, w):
------- self.edges.append((u, v, w))

--- def solve(self, root):
------- n = self.n
------- edges = self.edges
------- in_edge = [(-1, INF)] * n
------- for u, v, w in edges:
----------- if v == root:
--------------- continue
----------- if w < in_edge[v][1]:
--------------- in_edge[v] = (u, w)

------- in_edge[root] = (root, 0)
------- vis = [-1] * n
------- id_ = [-1] * n
------- cycle_id = 0

------- for i in range(n):
----------- v = i
----------- path = []

----------- while v != root and vis[v] == -1:
--------------- vis[v] = i
--------------- path.append(v)
--------------- v = in_edge[v][0]

--------------- if v == -1:
------------------- break

----------- if v != -1 and vis[v] == i:
--------------- while True:
------------------- u = path.pop()
------------------- id_[u] = cycle_id
------------------- if u == v:
----------------------- break
--------------- cycle_id += 1

------- if cycle_id == 0:
----------- return {v: in_edge[v] for v in range(n) if v != root}

------- for i in range(n):
----------- if id_[i] == -1:
--------------- id_[i] = cycle_id
--------------- cycle_id += 1

------- new_n = cycle_id
------- new_edges = []

------- for u, v, w in edges:
----------- nu, nv = id_[u], id_[v]
----------- if nu != nv:
--------------- new_edges.append((nu, nv, w - in_edge[v][1]))

------- contracted = Arborescence(new_n)
------- contracted.edges = new_edges

------- new_root = id_[root]
------- res = contracted.solve(new_root)

------- final = {}
------- for v in range(n):
----------- if v == root:
--------------- continue
----------- u, w = in_edge[v]
----------- final[v] = (u, w)

------- return final
```

5. Math I Algebra

5.0.1. *Generating Function Manager.*

```
MOD = 998244353

def mod_pow(base, exp):
--- res = 1
--- base %= MOD
--- while exp:
------- if exp & 1:
----------- res = res * base % MOD
------- base = base * base % MOD
------- exp >>= 1
--- return res

class GF_Manager:
--- DEPTH = 23

--- def __init__(self):
------- self.prim = [0] * (self.DEPTH + 1)
------- self.prim_inv = [0] * (self.DEPTH + 1)
------- self.two_inv = [0] * (self.DEPTH + 1)
------- self.set_up_primitives()

--- def set_up_primitives(self):
------- self.prim[self.DEPTH] = 31
------- self.prim_inv[self.DEPTH] = mod_pow(self.prim[self.DEPTH], MOD
↪  - 2)
------- self.two_inv[self.DEPTH] = mod_pow(1 << self.DEPTH, MOD - 2) -

------- for n in range(self.DEPTH - 1, -1, -1):
----------- self.prim[n] = self.prim[n + 1] * self.prim[n + 1] % MOD -
----------- self.prim_inv[n] = mod_pow(self.prim[n], MOD - 2)
----------- self.two_inv[n] = mod_pow(1 << n, MOD - 2)

--- def NTT(self, A, n, is_inverse=False):
------- if n == 0:
----------- return A[:]

------- half = 1 << (n - 1)
------- even = A[0::2]
------- odd = A[1::2]

------- even = self.NTT(even, n - 1, is_inverse)
------- odd = self.NTT(odd, n - 1, is_inverse)

------- res = [0] * (1 << n)
------- w1 = self.prim_inv[n] if is_inverse else self.prim[n]
------- w = 1

------- for i in range(half):
----------- t = w * odd[i] % MOD
----------- res[i] = (even[i] + t) % MOD
----------- res[i + half] = (even[i] - t) % MOD
----------- w = w * w1 % MOD

------- return res

--- def mult(self, A, B):
------- an, bn = len(A), len(B)
------- degree = an + bn - 1

------- n = 0
------- while (1 << n) < degree:
----------- n += 1

------- size = 1 << n
------- tA = A + [0] * (size - an)
------- tB = B + [0] * (size - bn)

------- tA = self.NTT(tA, n)
------- tB = self.NTT(tB, n)
```

```
------- for i in range(size):
----------- tA[i] = tA[i] * tB[i] % MOD

------- tA = self.NTT(tA, n, True)

------- inv = self.two_inv[n]
------- return [(x * inv) % MOD for x in tA[:degree]]

--- def horners(self, F, x):
------- ans = 0
------- for coef in reversed(F):
----------- ans = (ans * x + coef) % MOD
------- return ans
```

5.0.2. *Fast Fourier Transform.*

```
import math

class Poly:
--- def __init__(self, a=0.0, b=0.0):
------- self.a = a
------- self.b = b

--- def __add__(self, other):
------- return Poly(self.a + other.a, self.b + other.b)

--- def __sub__(self, other):
------- return Poly(self.a - other.a, self.b - other.b)

--- def __mul__(self, other):
------- return Poly(
----------- self.a * other.a - self.b * other.b,
----------- self.a * other.b + self.b * other.a
------- )

--- def __repr__(self):
------- return f"({self.a}+{self.b}i)"

def fft_recursive(inp, n, s):
--- if n == 1:
------- return [inp[0]]

--- half = n // 2

--- even = fft_recursive(inp, half, s * 2)
--- odd = fft_recursive(inp[s:], half, s * 2)

--- res = [None] * n
--- w = Poly(1, 0)
--- wn = Poly(math.cos(math.pi / half), math.sin(math.pi / half))

--- for i in range(half):
------- t = w * odd[i]
------- res[i] = even[i] + t
------- res[i + half] = even[i] - t
------- w = w * wn

--- return res

def fft(p):
--- return fft_recursive(p, len(p), 1)

def inverse_fft(p):
--- n = len(p)

--- p = [Poly(x.a, -x.b) for x in p]

--- p = fft(p)

--- res = []
--- for x in p:
------- res.append(Poly(x.a / n, -x.b / n))

--- return res
```

5.0.3. *Fft Polynomial Multiplication.*

```
def fft_multiply(a, b):
--- an, bn = len(a), len(b)
--- degree = an + bn - 1

--- n = 1
--- while n < degree:
------- n <<= 1

--- A = [Poly(x, 0) for x in a] + [Poly(0, 0)] * (n - an)
--- B = [Poly(x, 0) for x in b] + [Poly(0, 0)] * (n - bn)

--- A = fft(A)
--- B = fft(B)

--- for i in range(n):
------- A[i] = A[i] * B[i]

--- A = inverse_fft(A)

--- result = [int(round(A[i].a)) for i in range(degree)]
--- return result
```

5.0.4. *Number Theoretic Transform.*

```
MOD = 998244353
PRIMITIVE_ROOT = 3

def mod_pow(base, exp, mod=MOD):
--- res = 1
--- base %= mod
--- while exp:
------- if exp & 1:
----------- res = res * base % mod
------- base = base * base % mod
------- exp >>= 1
--- return res

class Num:
--- def __init__(self, x=0):
------- self.x = x % MOD

--- def __add__(self, other):
------- return Num(self.x + other.x)

--- def __sub__(self, other):
------- return Num(self.x - other.x)

--- def __mul__(self, other):
------- return Num(self.x * other.x)

--- def __truediv__(self, other):
------- return self * other.inv()

--- def inv(self):
------- return Num(mod_pow(self.x, MOD - 2))

--- def pow(self, p):
------- return Num(mod_pow(self.x, p))

--- def __repr__(self):
------- return str(self.x)

def ntt(a, invert=False):
--- n = len(a)

--- j = 0
--- for i in range(1, n):
------- bit = n >> 1
------- while j & bit:
----------- j ^= bit
----------- bit >>= 1
------- j |= bit
------- if i < j:
----------- a[i], a[j] = a[j], a[i]

--- length = 2
--- while length <= n:
------- wlen = mod_pow(PRIMITIVE_ROOT, (MOD - 1) // length)
------- if invert:
----------- wlen = mod_pow(wlen, MOD - 2)

------- for i in range(0, n, length):
----------- w = 1
----------- half = length // 2
----------- for j in range(i, i + half):
--------------- u = a[j]
--------------- v = Num(a[j + half].x * w % MOD)
--------------- a[j] = u + v
--------------- a[j + half] = u - v
--------------- w = w * wlen % MOD

------- length <<= 1

--- if invert:
------- inv_n = mod_pow(n, MOD - 2)
------- for i in range(n):
----------- a[i] = Num(a[i].x * inv_n % MOD)

def multiply(A, B):
--- n = 1
--- while n < len(A) + len(B):
------- n <<= 1

--- fa = [Num(x) for x in A] + [Num(0)] * (n - len(A))
--- fb = [Num(x) for x in B] + [Num(0)] * (n - len(B))

--- ntt(fa)
--- ntt(fb)

--- for i in range(n):
------- fa[i] = fa[i] * fb[i]

--- ntt(fa, True)

--- return [fa[i].x for i in range(len(A) + len(B) - 1)]

def poly_inv(x, l):
--- if l == 1:
------- return [mod_pow(x[0], MOD - 2)]

--- y = poly_inv(x, (l + 1) // 2)

--- n = 1
--- while n < 2 * l:
------- n <<= 1

--- A = x[:l] + [0] * (n - l)
--- B = y[:] + [0] * (n - len(y))

--- A = [Num(v) for v in A]
--- B = [Num(v) for v in B]

--- ntt(A)
--- ntt(B)

--- for i in range(n):
------- B[i] = B[i] * (Num(2) - A[i] * B[i])
```

```
--- ntt(B, True) ----------------------------------------------------------
-----------------------------------------------------------------------------
--- return [B[i].x % MOD for i in range(l)] --------------------------------
-----------------------------------------------------------------------------
def poly_sqrt(x, l): ---------------------------------------------------------
--- if l == 1: ------------------------------------------------------------
------- assert x[0] == 1 ----------------------------------------------------
------- return [1] ----------------------------------------------------------
-----------------------------------------------------------------------------
--- y = poly_sqrt(x, l // 2) ----------------------------------------------
--- inv_y = poly_inv(y, l // 2) -------------------------------------------
-----------------------------------------------------------------------------
--- n = 1 -----------------------------------------------------------------
--- while n < 2 * l: ------------------------------------------------------
------- n <<= 1 -------------------------------------------------------------
-----------------------------------------------------------------------------
--- T1 = x[:l] + [0] * (n - l) --------------------------------------------
--- T2 = inv_y[:] + [0] * (n - len(inv_y)) --------------------------------
-----------------------------------------------------------------------------
--- T1 = [Num(v) for v in T1] ---------------------------------------------
--- T2 = [Num(v) for v in T2] ---------------------------------------------
-----------------------------------------------------------------------------
--- ntt(T1) ---------------------------------------------------------------
--- ntt(T2) ---------------------------------------------------------------
-----------------------------------------------------------------------------
--- for i in range(n): ----------------------------------------------------
------- T2[i] = T1[i] * T2[i] -----------------------------------------------
-----------------------------------------------------------------------------
--- ntt(T2, True) ---------------------------------------------------------
-----------------------------------------------------------------------------
--- inv2 = (MOD + 1) // 2 -------------------------------------------------
--- res = [] --------------------------------------------------------------
--- for i in range(l): ----------------------------------------------------
------- val = (y[i] + T2[i].x) * inv2 % MOD ---------------------------------
------- res.append(val) -----------------------------------------------------
-----------------------------------------------------------------------------
--- return res ------------------------------------------------------------
```

5.0.5. *Polynomial Long Division.*

```
MOD = 998244353 -------------------------------------------------------------
G = 3 -----------------------------------------------------------------------
-----------------------------------------------------------------------------
def mod_pow(a, e): ----------------------------------------------------------
--- r = 1 -----------------------------------------------------------------
--- while e: --------------------------------------------------------------
------- if e & 1: -----------------------------------------------------------
----------- r = r * a % MOD -----------------------------------------------
------- a = a * a % MOD -----------------------------------------------------
------- e >>= 1 -------------------------------------------------------------
--- return r --------------------------------------------------------------
-----------------------------------------------------------------------------
def bit_reverse(a): ---------------------------------------------------------
--- n = len(a) ------------------------------------------------------------
--- j = 0 -----------------------------------------------------------------
--- for i in range(1, n): -------------------------------------------------
------- bit = n >> 1 --------------------------------------------------------
------- while j & bit: ------------------------------------------------------
----------- j ^= bit ------------------------------------------------------
----------- bit >>= 1 -----------------------------------------------------
------- j ^= bit ------------------------------------------------------------
------- if i < j: -----------------------------------------------------------
----------- a[i], a[j] = a[j], a[i] ---------------------------------------
-----------------------------------------------------------------------------
-----------------------------------------------------------------------------
def ntt(a, invert=False): ---------------------------------------------------
--- n = len(a) ------------------------------------------------------------
--- bit_reverse(a) --------------------------------------------------------
-----------------------------------------------------------------------------
--- length = 2 ------------------------------------------------------------
--- while length <= n: ----------------------------------------------------
------- wlen = mod_pow(G, (MOD - 1) // length) ------------------------------
------- if invert: ----------------------------------------------------------
----------- wlen = mod_pow(wlen, MOD - 2) -------------------------------
-----------------------------------------------------------------------------
------- for i in range(0, n, length): ---------------------------------------
----------- w = 1 -------------------------------------------------------
----------- half = length // 2 ------------------------------------------
----------- for j in range(i, i + half): --------------------------------
--------------- u = a[j] ----------------------------------------------
--------------- v = a[j + half] * w % MOD -----------------------------
--------------- a[j] = (u + v) % MOD ----------------------------------
--------------- a[j + half] = (u - v) % MOD ---------------------------
--------------- w = w * wlen % MOD ------------------------------------
-----------------------------------------------------------------------------
------- length <<= 1 --------------------------------------------------------
-----------------------------------------------------------------------------
--- if invert: ------------------------------------------------------------
------- inv_n = mod_pow(n, MOD - 2) -----------------------------------------
------- for i in range(n): --------------------------------------------------
----------- a[i] = a[i] * inv_n % MOD -------------------------------------
-----------------------------------------------------------------------------
def multiply(a, b): ---------------------------------------------------------
--- n = 1 -----------------------------------------------------------------
--- while n < len(a) + len(b): --------------------------------------------
------- n <<= 1 -------------------------------------------------------------
-----------------------------------------------------------------------------
--- fa = a[:] + [0] * (n - len(a)) ----------------------------------------
--- fb = b[:] + [0] * (n - len(b)) ----------------------------------------
-----------------------------------------------------------------------------
--- ntt(fa) ---------------------------------------------------------------
--- ntt(fb) ---------------------------------------------------------------
-----------------------------------------------------------------------------
--- for i in range(n): ----------------------------------------------------
------- fa[i] = fa[i] * fb[i] % MOD -----------------------------------------
-----------------------------------------------------------------------------
--- ntt(fa, True) ---------------------------------------------------------
-----------------------------------------------------------------------------
--- return fa[:len(a) + len(b) - 1] ----------------------------------------
-----------------------------------------------------------------------------
def poly_inv(f, n): ---------------------------------------------------------
--- if n == 1: ------------------------------------------------------------
------- return [mod_pow(f[0], MOD - 2)] --------------------------------------
-----------------------------------------------------------------------------
--- m = (n + 1) // 2 ------------------------------------------------------
--- g = poly_inv(f, m) ----------------------------------------------------
-----------------------------------------------------------------------------
--- size = 1 --------------------------------------------------------------
--- while size < 2 * n: ---------------------------------------------------
------- size <<= 1 ----------------------------------------------------------
-----------------------------------------------------------------------------
--- a = f[:n] + [0] * (size - len(f[:n])) ---------------------------------
--- g = g + [0] * (size - len(g)) -----------------------------------------
-----------------------------------------------------------------------------
--- ntt(a) ----------------------------------------------------------------
--- ntt(g) ----------------------------------------------------------------
-----------------------------------------------------------------------------
--- for i in range(size): -------------------------------------------------
------- g[i] = g[i] * (2 - a[i] * g[i] % MOD) % MOD -------------------------
-----------------------------------------------------------------------------
--- ntt(g, True) ----------------------------------------------------------
-----------------------------------------------------------------------------
--- return g[:n] ----------------------------------------------------------
-----------------------------------------------------------------------------
def poly_divide(A, B): ------------------------------------------------------
--- n, m = len(A), len(B) -------------------------------------------------
-----------------------------------------------------------------------------
--- if n < m: -------------------------------------------------------------
------- return [], A[:] -----------------------------------------------------
-----------------------------------------------------------------------------
--- k = n - m + 1 ---------------------------------------------------------
-----------------------------------------------------------------------------
--- A_rev = A[::-1] -------------------------------------------------------
--- B_rev = B[::-1] -------------------------------------------------------
-----------------------------------------------------------------------------
--- B_inv = poly_inv(B_rev, k) --------------------------------------------
-----------------------------------------------------------------------------
--- Q_rev = multiply(A_rev, B_inv)[:k] ------------------------------------
--- Q = Q_rev[::-1] -------------------------------------------------------
-----------------------------------------------------------------------------
--- BQ = multiply(B, Q) ---------------------------------------------------
-----------------------------------------------------------------------------
--- R = [(A[i] - (BQ[i] if i < len(BQ) else 0)) % MOD for i in
↪   range(n)]
-----------------------------------------------------------------------------
--- while Q and Q[-1] == 0: -----------------------------------------------
------- Q.pop() -------------------------------------------------------------
--- while R and R[-1] == 0: -----------------------------------------------
------- R.pop() -------------------------------------------------------------
-----------------------------------------------------------------------------
--- return Q, R -----------------------------------------------------------
```

5.0.6. *Matrix Multiplication.*

```
MOD = 998244353 -------------------------------------------------------------
-----------------------------------------------------------------------------
def multiply(A, B): ---------------------------------------------------------
--- p = len(A) ------------------------------------------------------------
--- q = len(A[0]) ---------------------------------------------------------
--- r = len(B[0]) ---------------------------------------------------------
-----------------------------------------------------------------------------
--- if len(B) != q: -------------------------------------------------------
------- raise ValueError("Dimension mismatch") ------------------------------
-----------------------------------------------------------------------------
--- AB = [[0] * r for _ in range(p)] --------------------------------------
-----------------------------------------------------------------------------
--- for i in range(p): ----------------------------------------------------
------- for j in range(q): --------------------------------------------------
----------- for k in range(r): ------------------------------------------
--------------- AB[i][k] = (AB[i][k] + A[i][j] * B[j][k]) % MOD ------
-----------------------------------------------------------------------------
--- return AB -------------------------------------------------------------
```

5.0.7. *Matrix Power.*

```
MOD = 998244353 -------------------------------------------------------------
-----------------------------------------------------------------------------
def multiply(A, B): ---------------------------------------------------------
--- p, q, r = len(A), len(A[0]), len(B[0]) --------------------------------
-----------------------------------------------------------------------------
--- res = [[0] * r for _ in range(p)] -------------------------------------
-----------------------------------------------------------------------------
--- for i in range(p): ----------------------------------------------------
------- for j in range(q): --------------------------------------------------
----------- for k in range(r): ------------------------------------------
--------------- res[i][k] = (res[i][k] + A[i][j] * B[j][k]) % MOD ----
-----------------------------------------------------------------------------
--- return res ------------------------------------------------------------
-----------------------------------------------------------------------------
def identity(n): ------------------------------------------------------------
--- I = [[0] * n for _ in range(n)] ---------------------------------------
--- for i in range(n): ----------------------------------------------------
------- I[i][i] = 1 ---------------------------------------------------------
--- return I --------------------------------------------------------------
-----------------------------------------------------------------------------
def power(B, e): ------------------------------------------------------------
--- n = len(B) ------------------------------------------------------------
-----------------------------------------------------------------------------
--- result = identity(n) --------------------------------------------------
--- base = [row[:] for row in B] ------------------------------------------
-----------------------------------------------------------------------------
--- while e > 0: ----------------------------------------------------------
------- if e & 1: -----------------------------------------------------------
----------- result = multiply(result, base) -------------------------------
------- base = multiply(base, base) -----------------------------------------
------- e >>= 1 -------------------------------------------------------------
-----------------------------------------------------------------------------
```

```
--- return result
```

5.0.8. *Gauss Jordan Matrix Determinant.*

```
EPS = 1e-9

def gauss_jordan(A):
--- n = len(A)
--- m = len(A[0])

--- singular = False

--- i = 0
--- for p in range(m):
------- if i >= n:
----------- break

------- pivot = i
------- for k in range(i + 1, n):
----------- if abs(A[k][p]) > abs(A[pivot][p]):
--------------- pivot = k

------- if abs(A[pivot][p]) < EPS:
----------- singular = True
----------- continue

------- A[i], A[pivot] = A[pivot], A[i]

------- div = A[i][p]
------- for j in range(p, m):
----------- A[i][j] /= div

------- for k in range(n):
----------- if k == i:
--------------- continue
----------- factor = A[k][p]
----------- for j in range(p, m):
--------------- A[k][j] -= factor * A[i][j]

------- i += 1

--- return not singular

def apply(A, x):
--- return [sum(A[i][j] * x[j] for j in range(len(x))) for i in
↪  range(len(A))]
```

6. Math II Combinatorics

6.0.1. *Lucas Theorem.*

```
_fact_cache = []
_last_p = -1

def lucas(n, k, p):
--- if k < 0 or k > n:
------- return 0
--- if k == 0 or n == k:
------- return 1

--- global _fact_cache, _last_p
--- if _last_p != p:
------- _fact_cache = [1] * p
------- for i in range(2, p):
----------- _fact_cache[i] = (_fact_cache[i - 1] * i) % p
------- _last_p = p

--- res_num = 1
--- res_den = 1

--- while n > 0 or k > 0:
------- ni, ki = n % p, k % p
------- if ki > ni:
----------- return 0
------- res_num = (res_num * _fact_cache[ni]) % p
------- res_den = (res_den * _fact_cache[ki] * _fact_cache[ni - ki]) %
↪  p
------- n //= p
------- k //= p
--- return (res_num * pow(res_den, p - 2, p)) % p
# Input Format: print(lucas(n, k, p))
```

6.0.2. *Catalan Numbers.*

```
def catalan(n):
--- if (n == 0 or n == 1):
------- return 1
---

--- dp = [0] * (n + 1)
--- dp[0] = dp[1] = 1

--- for i in range(2, n + 1):
------- for j in range(i):
----------- dp[i] += dp[j] * dp[i-j-1]

--- return dp[n]

# for i in range(10):
#     print(catalan(i))
```

6.0.3. *Kth Permutation.*

```
#fact = [1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880]

def findKthPermutation(n, k):
--- nums = [str(i) for i in range(1, n + 1)]

--- fact = [1] * n
--- for i in range(1, n):
------- fact[i] = fact[i - 1] * i

--- k -= 1
--- res = []

--- for i in range(n - 1, -1, -1):
------- idx = k // fact[i]
------- k %= fact[i]

------- res.append(nums.pop(idx))

--- return ''.join(res)
```

6.0.4. *Factoradics.*

```
def factoradic(arr):
--- n = len(arr)
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in arr:
------- bit_val = val + 1
------- s = ft.query(bit_val - 1)
------- ft.add(bit_val, -1)
------- res.append(s)
--- return res

def permute(fact, n):
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in fact:
------- idx = ft.find_kth(val)
------- ft.add(idx, -1)
------- res.append(idx - 1)
--- return res
```

7. Math III Number Theory

7.0.1. *Gcd Subset Counter.*

```
def gcd_subset_count(arr):
--- n = len(arr)
--- mx = max(arr)

--- f = [0] * (mx + 1)
--- for x in arr:
------- f[x] += 1

--- gcnt = [0] * (mx + 1)

--- for i in range(mx, 0, -1):
------- add = 0

------- for j in range(i, mx + 1, i):
----------- add += f[j]

------- total = (1 << add) - 1  # 2^add - 1

------- sub = 0
------- for j in range(2 * i, mx + 1, i):
----------- sub += gcnt[j]

------- gcnt[i] = total - sub

--- return gcnt
```

7.0.2. *Linear Diophantine.*

```
def gcd(a, b):
--- if a == 0:
------- return b
--- return gcd(b % a, a)

def isPossible(a, b, c):
--- g = gcd(a, b)

--- return c % g == 0
```

7.0.3. *Chinese Remainder Theorem.*

```
def gcd_extended(a, b):
--- if a == 0:
------- return b, 0, 1
--- gcd, x1, y1 = gcd_extended(b % a, a)
--- x = y1 - (b // a) * x1
--- y = x1
--- return gcd, x, y

def find_min_x(num, rem):
--- prod = 1
--- for n in num:
------- prod *= n

--- result = 0
--- for i in range(len(num)):
------- prod_i = prod // num[i]
------- _, inv_i, _ = gcd_extended(prod_i, num[i])
------- result += rem[i] * prod_i * inv_i

--- return result % prod
```

7.0.4. *Euler Totient.*

```
def totient(n):
--- if n <= 1:
------- return 1

--- tot = n
--- i = 2

--- while i * i <= n:
------- if n % i == 0:
----------- tot -= tot // i
----------- while n % i == 0:
--------------- n //= i
------- i += 1

--- if n > 1:
------- tot -= tot // n

--- return tot
```

7.0.5. *Extended Euclidean Algorithm.*

```
def mod(x, m):
--- if m == 0:
------- return 0
--- if m < 0:
------- m = -m
--- return (x % m + m) % m


def extended_euclid(a, b):
--- if b == 0:
------- return a, 1, 0

--- g, x1, y1 = extended_euclid(b, a % b)

--- x = y1
--- y = x1 - (a // b) * y1

--- return g, x, y
```

7.0.6. *Modular Exponentiation.*

```
def mod_pow(b, e, m):
--- res = 1 % m

--- while e > 0:
------- if e & 1:
----------- res = (res * b) % m
------- b = (b * b) % m
------- e >>= 1

--- return res
```

7.0.7. *Mod Inverse.*

```
def extended_euclid(a, b):
--- if b == 0:
------- return a, 1, 0

--- g, x1, y1 = extended_euclid(b, a % b)

--- x = y1
--- y = x1 - (a // b) * y1

--- return g, x, y


def modinv(a, m):
--- g, x, y = extended_euclid(a, m)

--- if g != 1:
------- return 0  # no inverse

--- return x % m
```

7.0.8. *Modulo Solver.*

```
def extended_euclid(a, b):
--- if b == 0:
------- return a, 1, 0

--- g, x1, y1 = extended_euclid(b, a % b)

--- x = y1
--- y = x1 - (a // b) * y1

--- return g, x, y


def modsolver(a, b, m):
--- g, x, y = extended_euclid(a, m)

--- if b % g != 0:
------- return (-1, -1)

--- m_reduced = m // g

--- x0 = (x * (b // g)) % m_reduced

--- return (x0, m_reduced)
```

7.0.9. *Primitive Root.*

```
def primitive_root(m):
--- if m == 2:
------- return 1

--- phi = m - 1

--- factors = set()
--- x = phi
--- i = 2

--- while i * i <= x:
------- if x % i == 0:
----------- factors.add(i)
----------- while x % i == 0:
--------------- x //= i
------- i += 1

--- if x > 1:
------- factors.add(x)

--- for g in range(2, m):
------- ok = True
------- for f in factors:
----------- if mod_pow(g, phi // f, m) == 1:
--------------- ok = False
--------------- break
------- if ok:
----------- return g

--- return -1
```

7.0.10. *Josephus.*

```
def josephus(n, k):
--- res = 0
--- for i in range(2, n + 1):
------- res = (res + k) % i
--- return res
```

8. Math IV Numerical Methods

8.0.1. *Fast Square Testing.*

```
import math


class FastSquareTester:
--- def __init__(self):
------- self.M = 0
------- self.init_is_square()

--- def init_is_square(self):
------- for i in range(64):
----------- self.M |= (1 << (i * i % 64))

--- def is_square(self, x):
------- if x < 0: return False
------- if x == 0: return True

------- if not (self.M & (1 << (x & 63))):
----------- return False

------- c = (x & -x).bit_length() - 1
------- if c & 1:
----------- return False

------- temp_x = x >> c
------- if (temp_x & 7) != 1:
----------- return False

------- r = int(math.isqrt(temp_x))
------- return r * r == temp_x

tester = FastSquareTester()
```

8.0.2. *Simpson Integration.*

```
def f(x):
--- return x ** 2
def simpson_integration(a, b, n=1000000):
--- if n % 2 != 0:
------- n += 1

--- h = (b - a) / n
--- s = f(a) + f(b)

--- for i in range(1, n):
------- x = a + h * i
------- if i % 2 == 1:
----------- s += 4 * f(x)
------- else:
----------- s += 2 * f(x)

--- return s * (h / 3)
```

9. Strings

9.0.1. *Longest Common Prefix.*

```
def build_lcp(s, sa):
--- n = len(s)
--- rank = [0] * n
--- lcp = [0] * n

--- for i in range(n):
------- rank[sa[i]] = i

--- k = 0
--- for i in range(n):
------- if rank[i] == 0:
----------- k = 0
----------- continue
```

```
------- j = sa[rank[i] - 1] ----------------------------------------
--------------------------------------------------------------------
------- while i + k < n and j + k < n and s[i + k] == s[j + k]: ------
----------- k += 1 -------------------------------------------------
--------------------------------------------------------------------
------- lcp[rank[i]] = k ---------------------------------------------
--------------------------------------------------------------------
------- if k > 0: ----------------------------------------------------
----------- k -= 1 -------------------------------------------------
--------------------------------------------------------------------
--- return lcp -----------------------------------------------------
```

9.0.2. *Hasher.*

```
class Hasher: --------------------------------------------------------
--- def __init__(self, s, bases=[31, 37], mod=10 ** 9 + 7): ---------
------- self.bases = bases -------------------------------------------
------- self.mod = mod -----------------------------------------------
------- self.n = len(s) ----------------------------------------------
------- self.num_hashes = len(bases) ---------------------------------
------- self.p_pow = [[1] * (self.n + 1) for _ in
↪  range(self.num_hashes)]
------- self.h_ans = [[0] * (self.n + 1) for _ in
↪  range(self.num_hashes)]
--------------------------------------------------------------------
------- for i, b in enumerate(self.bases): ---------------------------
----------- for j in range(self.n): ------------------------------------
--------------- self.p_pow[i][j + 1] = (self.p_pow[i][j] * b) %
↪  self.mod
--------------- self.h_ans[i][j + 1] = (self.h_ans[i][j] + s[j] *
↪  self.p_pow[i][j]) % self.mod
--------------------------------------------------------------------
--- def get_hash(self, l, r): --------------------------------------
------- return tuple((self.h_ans[i][r + 1] - self.h_ans[i][l]) %
↪  self.mod
-------------------- for i in range(self.num_hashes)) ----------------
--------------------------------------------------------------------
--- def are_equal(self, l1, r1, l2, r2): ---------------------------
------- if (r1 - l1) != (r2 - l2): return False ----------------------
--------------------------------------------------------------------
------- h1 = self.get_hash(l1, r1) -----------------------------------
------- h2 = self.get_hash(l2, r2) -----------------------------------
--------------------------------------------------------------------
------- return all((h1[i] * self.p_pow[i][l2]) % self.mod == ---------
----------------- (h2[i] * self.p_pow[i][l1]) % self.mod ------------
----------------- for i in range(self.num_hashes)) ------------------
```

9.0.3. *Z Algorithm.*

9.0.4. *Aho Corasick Trie.*

```
from collections import deque ----------------------------------------
import random --------------------------------------------------------
import string --------------------------------------------------------
--------------------------------------------------------------------
class Node: ----------------------------------------------------------
--- def __init__(self): --------------------------------------------
------- self.next = {} -----------------------------------------------
------- self.fail = None ---------------------------------------------
------- self.count = 0 -----------------------------------------------
--------------------------------------------------------------------
--- def add(self, s): ----------------------------------------------
------- node = self --------------------------------------------------
------- for c in s: --------------------------------------------------
----------- if c not in node.next: -------------------------------------
--------------- node.next[c] = Node() --------------------------------
---------- node = node.next[c] -----------------------------------------
------- node.count += 1 ----------------------------------------------
--------------------------------------------------------------------
--- def prepare(self): ---------------------------------------------
------- root = self --------------------------------------------------
------- q = deque() --------------------------------------------------
--------------------------------------------------------------------
------- root.fail = root ---------------------------------------------
--------------------------------------------------------------------
------- for child in root.next.values(): -----------------------------
---------- child.fail = root -------------------------------------------
---------- q.append(child) ---------------------------------------------
--------------------------------------------------------------------
------- while q: -----------------------------------------------------
---------- head = q.popleft() ------------------------------------------
--------------------------------------------------------------------
---------- for c, nxt in head.next.items(): ----------------------------
--------------- f = head.fail ----------------------------------------
--------------------------------------------------------------------
--------------- while f != root and c not in f.next: -----------------
------------------ f = f.fail -------------------------------------
--------------------------------------------------------------------
--------------- nxt.fail = f.next[c] if c in f.next else root --------
--------------------------------------------------------------------
--------------- nxt.count += nxt.fail.count --------------------------
--------------------------------------------------------------------
--------------- q.append(nxt) ----------------------------------------
--------------------------------------------------------------------
--- def search(self, s): -------------------------------------------
------- root = self --------------------------------------------------
------- p = self -----------------------------------------------------
------- ans = 0 ------------------------------------------------------
--------------------------------------------------------------------
------- for c in s: --------------------------------------------------
---------- while p != root and c not in p.next: ------------------------
--------------- p = p.fail -------------------------------------------
--------------------------------------------------------------------
---------- if c in p.next: ---------------------------------------------
--------------- p = p.next[c] ----------------------------------------
--------------- ans += p.count ---------------------------------------
--------------------------------------------------------------------
------- return ans ---------------------------------------------------
```

9.0.5. *Knuth Morris Pratt.*

```
def build_kmp(f): ----------------------------------------------------
--- n = len(f) -----------------------------------------------------
--- par = [0] * (n + 1) --------------------------------------------
--------------------------------------------------------------------
--- par[0] = -1 ----------------------------------------------------
--- if n > 0: ------------------------------------------------------
------- par[1] = 0 ---------------------------------------------------
--------------------------------------------------------------------
--- i, j = 2, 0 ----------------------------------------------------
--------------------------------------------------------------------
--- while i <= n: --------------------------------------------------
------- if f[i - 1] == f[j]: -----------------------------------------
----------- j += 1 -----------------------------------------------------
----------- par[i] = j -------------------------------------------------
----------- i += 1 -----------------------------------------------------
------- elif j > 0: --------------------------------------------------
----------- j = par[j] -------------------------------------------------
------- else: --------------------------------------------------------
----------- par[i] = 0 -------------------------------------------------
----------- i += 1 -----------------------------------------------------
--------------------------------------------------------------------
--- return par -----------------------------------------------------
--------------------------------------------------------------------
--------------------------------------------------------------------
def kmp(s, f): -------------------------------------------------------
--- par = build_kmp(f) ---------------------------------------------
--------------------------------------------------------------------
--- i = 0 ----------------------------------------------------------
--- j = 0 ----------------------------------------------------------
--- ans = [] -------------------------------------------------------
--------------------------------------------------------------------
--- n, m = len(s), len(f) ------------------------------------------
--------------------------------------------------------------------
--- while i + j < n: -----------------------------------------------
------- if s[i + j] == f[j]: -----------------------------------------
----------- j += 1 -----------------------------------------------------
----------- if j == m: -------------------------------------------------
--------------- ans.append(i) ----------------------------------------
--------------- i += j - par[j] --------------------------------------
--------------- if j > 0: --------------------------------------------
------------------ j = par[j] -------------------------------------
------- else: --------------------------------------------------------
----------- i += j - par[j] --------------------------------------------
----------- if j > 0: --------------------------------------------------
--------------- j = par[j] -------------------------------------------
--------------------------------------------------------------------
--- return ans -----------------------------------------------------
```

9.0.6. *Trie.*

```
class Node: ----------------------------------------------------------
--- def __init__(self): --------------------------------------------
------- self.children = {} -------------------------------------------
------- self.prefixes = 0 --------------------------------------------
------- self.words = 0 -----------------------------------------------
--------------------------------------------------------------------
class Trie: ----------------------------------------------------------
--- def __init__(self): --------------------------------------------
------- self.root = Node() -------------------------------------------
--------------------------------------------------------------------
--- def insert(self, seq): -----------------------------------------
------- cur = self.root ----------------------------------------------
--------------------------------------------------------------------
------- for i in range(len(seq) + 1): --------------------------------
----------- cur.prefixes += 1 ------------------------------------------
--------------------------------------------------------------------
----------- if i == len(seq): ------------------------------------------
--------------- cur.words += 1 ---------------------------------------
--------------- break ------------------------------------------------
--------------------------------------------------------------------
----------- head = seq[i] ----------------------------------------------
--------------------------------------------------------------------
----------- if head not in cur.children: -------------------------------
--------------- cur.children[head] = Node() --------------------------
--------------------------------------------------------------------
----------- cur = cur.children[head] -----------------------------------
--------------------------------------------------------------------
--- def countMatches(self, seq): -----------------------------------
------- cur = self.root ----------------------------------------------
--------------------------------------------------------------------
------- for c in seq: ------------------------------------------------
----------- if c not in cur.children: ----------------------------------
--------------- return 0 --------------------------------------------
----------- cur = cur.children[c] --------------------------------------
--------------------------------------------------------------------
------- return cur.words ---------------------------------------------
--------------------------------------------------------------------
--- def countPrefixes(self, seq): ----------------------------------
------- cur = self.root ----------------------------------------------
--------------------------------------------------------------------
------- for c in seq: ------------------------------------------------
----------- if c not in cur.children: ----------------------------------
--------------- return 0 --------------------------------------------
----------- cur = cur.children[c] --------------------------------------
--------------------------------------------------------------------
------- return cur.prefixes ------------------------------------------
```

9.0.7. *Suffix Array.*

```
class SuffixArray:
    def __init__(self):
        self.n = 0
        self.suffix = []
        self.equiv = []
        self.equiv_pair = []
        self.s = ""

    def build(self, s):
        if not s.endswith('$'):
            s += '$'

        self.s = s
        self.n = len(s)

        self.suffix = list(range(self.n))
        self.equiv = [0] * self.n
        self.equiv_pair = [(0, 0)] * self.n

        self.suffix.sort(key=lambda i: s[i])

        cls = 0
        for i in range(self.n):
            if i == 0 or s[self.suffix[i]] != s[self.suffix[i - 1]]:
                cls += 1
            self.equiv[self.suffix[i]] = cls

        k = 1
        while k < self.n:
            for i in range(self.n):
                self.equiv_pair[i] = (
                    self.equiv[i],
                    self.equiv[(i + k) % self.n]
                )

            self.suffix.sort(key=lambda i: self.equiv_pair[i])

            cls = 0
            new_equiv = [0] * self.n

            for i in range(self.n):
                if i == 0 or self.equiv_pair[self.suffix[i]] !=
↪   self.equiv_pair[self.suffix[i - 1]]:
                    cls += 1
                new_equiv[self.suffix[i]] = cls

            self.equiv = new_equiv
            k <<= 1

    def _cmp(self, start, pattern):
        s = self.s
        n = len(s)

        i = start
        j = 0

        while i < n and j < len(pattern):
            if s[i] != pattern[j]:
                return ord(s[i]) - ord(pattern[j])
            i += 1
            j += 1

        if j == len(pattern):
            return 0
        return -1

    def lower_bound(self, pattern):
        l, r = 0, self.n - 1
        res = self.n

        while l <= r:
            m = (l + r) // 2
            if self._cmp(self.suffix[m], pattern) >= 0:
                res = m
                r = m - 1
            else:
                l = m + 1

        return res

    def upper_bound(self, pattern):
        l, r = 0, self.n - 1
        res = self.n

        while l <= r:
            m = (l + r) // 2
            if self._cmp(self.suffix[m], pattern) > 0:
                res = m
                r = m - 1
            else:
                l = m + 1

        return res

    def count_occurrences(self, pattern):
        L = self.lower_bound(pattern)
        R = self.upper_bound(pattern)
        return max(0, R - L)
```

9.0.8. *Booth Min String Rotation.*

```
def booth(s: str) -> int:
    s = s + s
    n = len(s)

    f = [-1] * n
    k = 0

    for j in range(1, n):
        i = f[j - k - 1]

        while i != -1 and s[j] != s[k + i + 1]:
            if s[j] < s[k + i + 1]:
                k = j - i - 1
            i = f[i]

        if i == -1 and s[j] != s[k]:
            if s[j] < s[k]:
                k = j
            f[j - k] = -1
        else:
            f[j - k] = i + 1

    return k
```

10. Other Algorithms

10.0.1. *Longest Increasing Subsequence.*

```
def lis(arr):
    if not arr:
        return []

    n = len(arr)
    seq = []
    back = [0] * n

    for i in range(n):
        lo, hi = 0, len(seq)
        while lo < hi:
            mid = (lo + hi) // 2
            if arr[seq[mid]] < arr[i]:
                lo = mid + 1
            else:
                hi = mid

        res = lo
        if res < len(seq):
            seq[res] = i
        else:
            seq.append(i)
        back[i] = -1 if res == 0 else seq[res - 1]

    ans = []
    at = seq[-1]
    while at != -1:
        ans.append(at)
        at = back[at]

    return ans[::-1]
```

10.0.2. *Cycle Find.*

```
def find_cycle(x0, f):
    t = f(x0)
    h = f(f(x0))

    while t != h:
        t = f(t)
        h = f(f(h))

    mu = 0
    h = x0
    while t != h:
        t = f(t)
        h = f(h)
        mu += 1

    lam = 1
    h = f(t)
    while t != h:
        h = f(h)
        lam += 1

    return mu, lam
```

10.0.3. *2Sat.*

```
class TwoSAT:
    def __init__(self, n):
        self.n = n
        self.N = 2 * n
        self.adj = [[] for _ in range(self.N)]

    def var(self, x):
        return x << 1

    def neg(self, x):
        return x ^ 1

    def add_imp(self, x, y):
        self.adj[x].append(y)

    def add_or(self, x, y):
        self.add_imp(self.neg(x), y)
        self.add_imp(self.neg(y), x)

    def sat(self):
        n = self.N
        adj = self.adj

        index = 0
        stack = []
        onstack = [False] * n
        ids = [-1] * n
```

```
------- low = [0] * n
------- comp = [-1] * n
------- self.time = 0
------- self.comp_id = 0

------- def dfs(u):
----------- nonlocal index

----------- ids[u] = low[u] = index
----------- index += 1
----------- stack.append(u)
----------- onstack[u] = True

----------- for v in adj[u]:
--------------- if ids[v] == -1:
------------------- dfs(v)
------------------- low[u] = min(low[u], low[v])
--------------- elif onstack[v]:
------------------- low[u] = min(low[u], ids[v])

----------- if low[u] == ids[u]:
--------------- while True:
------------------- v = stack.pop()
------------------- onstack[v] = False
------------------- comp[v] = self.comp_id
------------------- if v == u:
----------------------- break
--------------- self.comp_id += 1

------- for i in range(n):
----------- if ids[i] == -1:
--------------- dfs(i)

------- for i in range(0, n, 2):
----------- if comp[i] == comp[i ^ 1]:
--------------- return False

------- return True
```

10.0.4. *Dpll.*

```
class SAT:
--- def __init__(self):
------- self.clauses = []
------- self.n = 0

--- def var(self):
------- self.n += 1
------- return self.n

--- def clause(self, c):
------- self.clauses.append(c)

--- def clause_sat(self, clause, assign):
------- unknown = False

------- for x in clause:
----------- v = abs(x)
----------- val = assign[v]

----------- if x > 0 and val == 1:
--------------- return True
----------- if x < 0 and val == 0:
--------------- return True

----------- if val == -1:
--------------- unknown = True

------- return False if not unknown else None

--- def propagate(self, assign):
------- changed = True

------- while changed:
----------- changed = False

----------- for clause in self.clauses:
--------------- sat = False
--------------- unassigned = []

--------------- for x in clause:
------------------- v = abs(x)

------------------- if x > 0 and assign[v] == 1:
----------------------- sat = True
----------------------- break
------------------- if x < 0 and assign[v] == 0:
----------------------- sat = True
----------------------- break

------------------- if assign[v] == -1:
----------------------- unassigned.append(x)

--------------- if sat:
------------------- continue

--------------- if len(unassigned) == 0:
------------------- return False  # contradiction

--------------- if len(unassigned) == 1:
------------------- x = unassigned[0]
------------------- v = abs(x)
------------------- val = 1 if x > 0 else 0

------------------- if assign[v] != -1 and assign[v] != val:
----------------------- return False

------------------- if assign[v] == -1:
----------------------- assign[v] = val
----------------------- changed = True

------- return True

--- def solve_dfs(self, assign, idx=1):
------- if idx > self.n:
----------- return True

------- if assign[idx] != -1:
----------- return self.solve_dfs(assign, idx + 1)

------- for val in [0, 1]:
----------- new_assign = assign[:]
----------- new_assign[idx] = val

----------- if self.propagate(new_assign):
--------------- if self.solve_dfs(new_assign, idx + 1):
------------------- return True

------- return False

--- def solve(self):
------- assign = [-1] * (self.n + 1)

------- if not self.propagate(assign):
----------- return False

------- return self.solve_dfs(assign)

--- def get_value(self, x):
------- return self.assign[x] == 1
```

10.0.5. *Stable Marriage.*

```
from collections import deque

def stable_marriage(n, m_pref, w_pref):
--- w_rank = [[0] * n for _ in range(n)]

--- for w in range(n):
------- for i, man in enumerate(w_pref[w]):
----------- w_rank[w][man] = i

--- q = deque(range(n))

--- at = [0] * n

--- eng = [-1] * n

--- res = [-1] * n

--- while q:
------- m = q.popleft()

------- while at[m] < n:
----------- w = m_pref[m][at[m]]
----------- at[m] += 1

----------- if eng[w] == -1:
--------------- eng[w] = m
--------------- res[m] = w
--------------- break

----------- if w_rank[w][m] < w_rank[w][eng[w]]:
--------------- q.append(eng[w])
--------------- eng[w] = m
--------------- res[m] = w
--------------- break

--- return res
```

10.0.6. *Dates.*

```
def c_div(a, b):
--- return int(a / b)

def date_to_int(y, m, d):
--- k = c_div(m - 14, 12)

--- return (
------- 1461 * (y + 4800 + k) // 4
------- + 367 * (m - 2 - k * 12) // 12
------- - 3 * ((y + 4900 + k) // 100) // 4
------- + d - 32075
--- )

def int_to_day(jd):
--- return jd % 7

def int_to_date(jd):
--- x = jd + 68569
--- n = 4 * x // 146097
--- x -= (146097 * n + 3) // 4

--- i = (4000 * (x + 1)) // 1461001
--- x -= 1461 * i // 4 - 31

--- j = 80 * x // 2447
--- d = x - 2447 * j // 80

--- x = j // 11
--- m = j + 2 - 12 * x
--- y = 100 * (n - 49) + i + x

--- return y, m, d
```

10.0.7. *Simulated Annealing.*

```
import random
import math
import time


def simulated_annealing(n, seconds):
    rng = random.Random()

    sol = list(range(1, n + 1))
    rng.shuffle(sol)

    def score(arr):
        return sum(abs(arr[i] - arr[i - 1]) for i in range(1, n))

    cur = score(sol)

    T0 = 100.0
    T1 = 0.001
    start = time.time()
    iters = 0

    while True:
        if iters % 16 == 0:
            progress = (time.time() - start) / seconds
            if progress >= 1.0:
                break
            temp = T0 * (T1 / T0) ** progress

        a = rng.randint(0, n - 2)

        x, y = sol[a], sol[a + 1]

        delta = 0

        if a > 0:
            delta += abs(sol[a - 1] - y) - abs(sol[a - 1] - x)

        if a + 2 < n:
            delta += abs(x - sol[a + 2]) - abs(y - sol[a + 2])

        accept = delta <= 0 or rng.random() < math.exp(-delta / temp)

        if accept:
            sol[a], sol[a + 1] = sol[a + 1], sol[a]
            cur += delta

        iters += 1

    return cur, sol
```

10.0.8. *Simplex.*

```
import math

EPS = 1e-9

class LPSolver:
    def __init__(self, A, b, c):
        self.m = len(b)
        self.n = len(c)
        self.N = [0] * (self.n + 1)
        self.B = [0] * self.m
        self.D = [[0.0] * (self.n + 2) for _ in range(self.m + 2)]

        for i in range(self.m):
            for j in range(self.n):
                self.D[i][j] = A[i][j]

        for i in range(self.m):
            self.B[i] = self.n + i
            self.D[i][self.n] = -1
            self.D[i][self.n + 1] = b[i]

        for j in range(self.n):
            self.N[j] = j
            self.D[self.m][j] = -c[j]

        self.N[self.n] = -1
        self.D[self.m + 1][self.n] = 1

    def pivot(self, r, s):
        inv = 1.0 / self.D[r][s]

        for i in range(self.m + 2):
            if i != r:
                for j in range(self.n + 2):
                    if j != s:
                        self.D[i][j] -= self.D[r][j] * self.D[i][s] * inv

        for j in range(self.n + 2):
            if j != s:
                self.D[r][j] *= inv

        for i in range(self.m + 2):
            if i != r:
                self.D[i][s] *= -inv

        self.D[r][s] = inv
        self.B[r], self.N[s] = self.N[s], self.B[r]

    def simplex(self, phase):
        x = self.m + 1 if phase == 1 else self.m

        while True:
            s = -1
            for j in range(self.n + 1):
                if phase == 2 and self.N[j] == -1:
                    continue
                if (s == -1 or
                    self.D[x][j] < self.D[x][s] or
                    (abs(self.D[x][j] - self.D[x][s]) < EPS and self.N[j] < self.N[s])):
                    s = j

            if self.D[x][s] > -EPS:
                return True

            r = -1
            for i in range(self.m):
                if self.D[i][s] < EPS:
                    continue
                if (r == -1 or
                    self.D[i][self.n + 1] / self.D[i][s] <
                    self.D[r][self.n + 1] / self.D[r][s] or
                    (abs(self.D[i][self.n + 1] / self.D[i][s] -
                         self.D[r][self.n + 1] / self.D[r][s]) < EPS and
                     self.B[i] < self.B[r])):
                    r = i

            if r == -1:
                return False

            self.pivot(r, s)

    def solve(self):
        x = [0.0] * self.n

        r = 0
        for i in range(1, self.m):
            if self.D[i][self.n + 1] < self.D[r][self.n + 1]:
                r = i

        if self.D[r][self.n + 1] < -EPS:
            self.pivot(r, self.n)
            if not self.simplex(1) or self.D[self.m + 1][self.n + 1] < -EPS:
                return -math.inf, x

        for i in range(self.m):
            if self.B[i] == -1:
                s = -1
                for j in range(self.n + 1):
                    if (s == -1 or
                        self.D[i][j] < self.D[i][s] or
                        (abs(self.D[i][j] - self.D[i][s]) < EPS and self.N[j] < self.N[s])):
                        s = j
                self.pivot(i, s)

        if not self.simplex(2):
            return math.inf, x

        for i in range(self.m):
            if self.B[i] < self.n:
                x[self.B[i]] = self.D[i][self.n + 1]

        return self.D[self.m][self.n + 1], x
```

10.0.9. *Fast Input Reading.*

```
import sys


data = sys.stdin.buffer.read().split()
it = iter(data)

def next_int():
    return int(next(it))
```

10.0.10. *Bit Hacks.*

```
def snoob(x):
    y = x & -x
    z = x + y
    return z | (((x ^ z) // y) >> 2)
```